2020 MathWorks
中国汽车年会

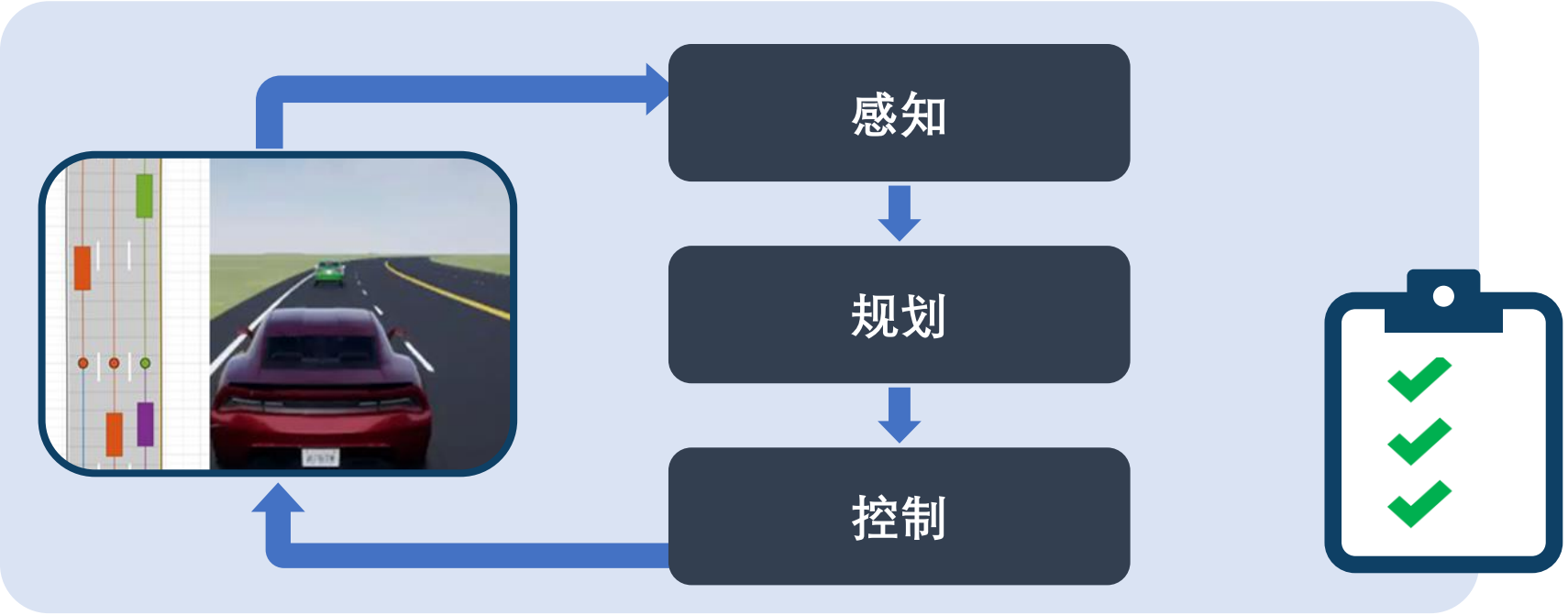
MATLAB 和 Simulink 用于开发自动驾驶的新特性

王鸿钧
MathWorks 中国区资深应用工程师

MathWorks®



自动驾驶工程师的几个常见问题



如何
分析 & 合成
场景?

如何
设计 & 部署
算法?

如何
集成 & 测试
系统?

分析与合成驾驶场景

分析实车数据的工作流程

数据获取

数据可视化

数据标注

开环仿真

合成虚拟场景的工作流程

创建环境场景

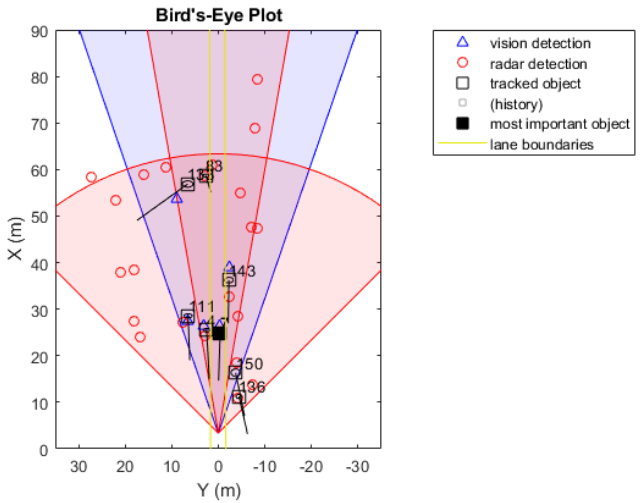
添加交通参与者

添加传感器

开环或闭环仿真

可视化多种自动驾驶数据

传感器检测

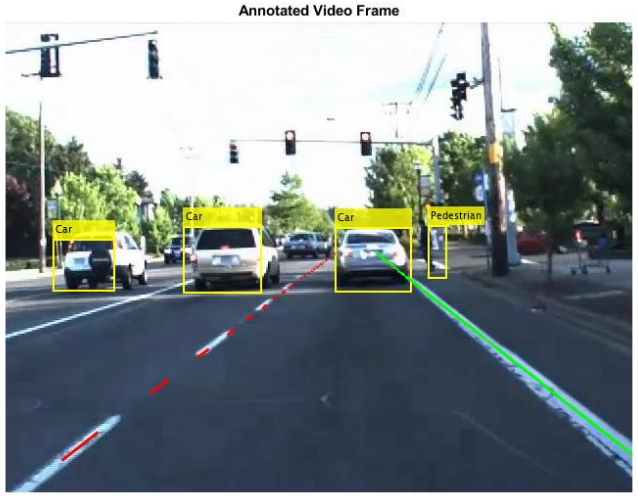


[Visualize Sensor Coverage, Detections, and Tracks](#)

Automated Driving Toolbox™

R2017a

图像

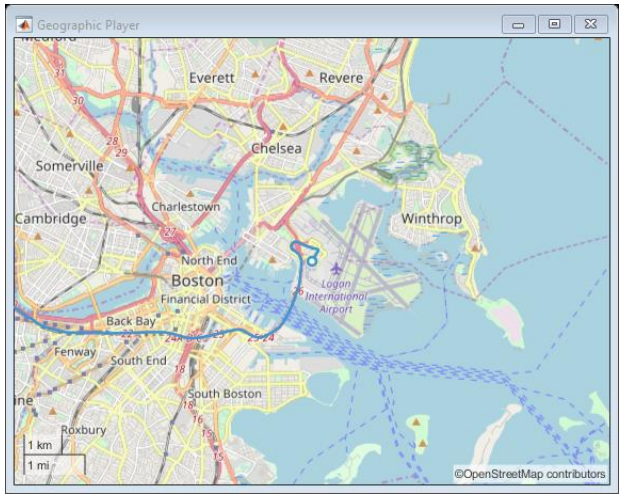


[Annotate Video Using Detections in Vehicle Coordinates](#)

Automated Driving Toolbox™

R2017a

地图



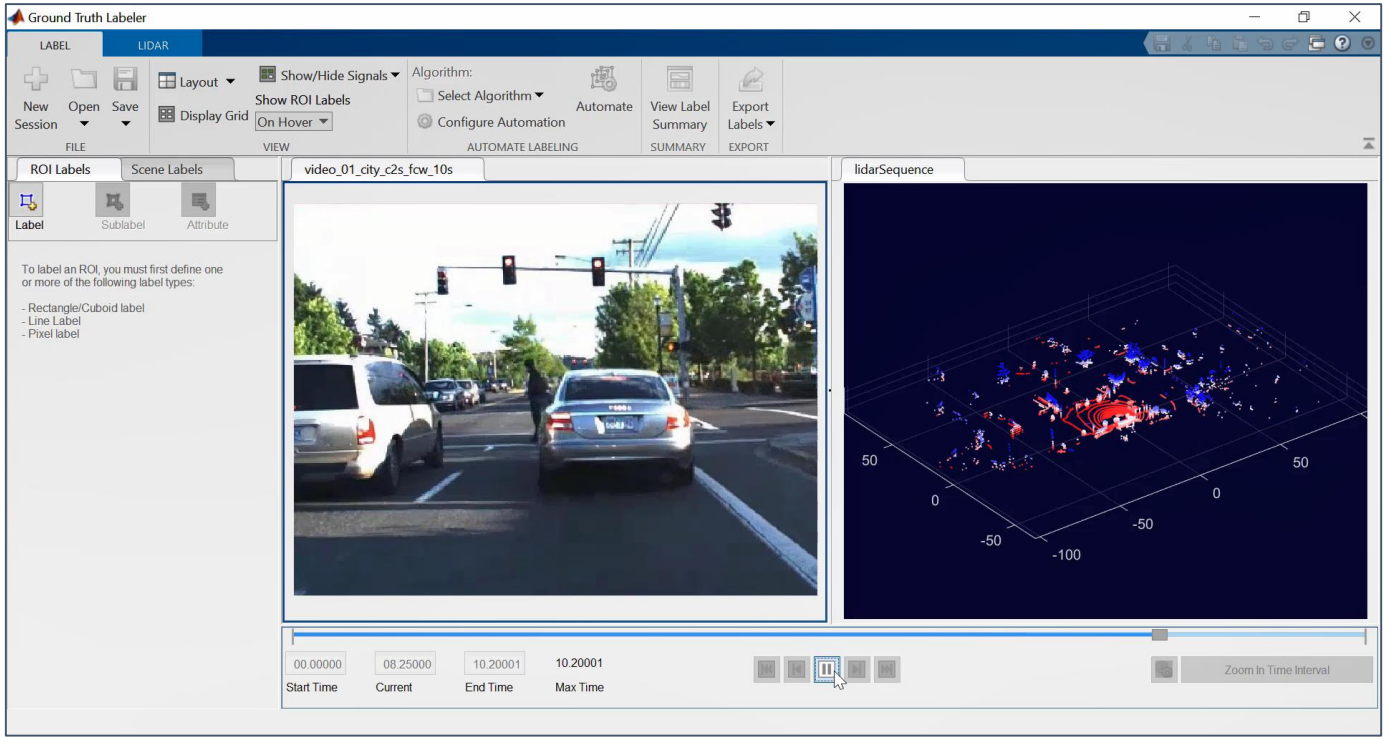
[Display Data on OpenStreetMap Basemap](#)

Automated Driving Toolbox™

R2018a

可视化和标注图像与激光雷达数据

多传感器可视化 → 交互式标注 → 自动标注 → 导出标签



- 载入时间上重叠的表达同一环境的多个传感器信号
- 同步浏览数据

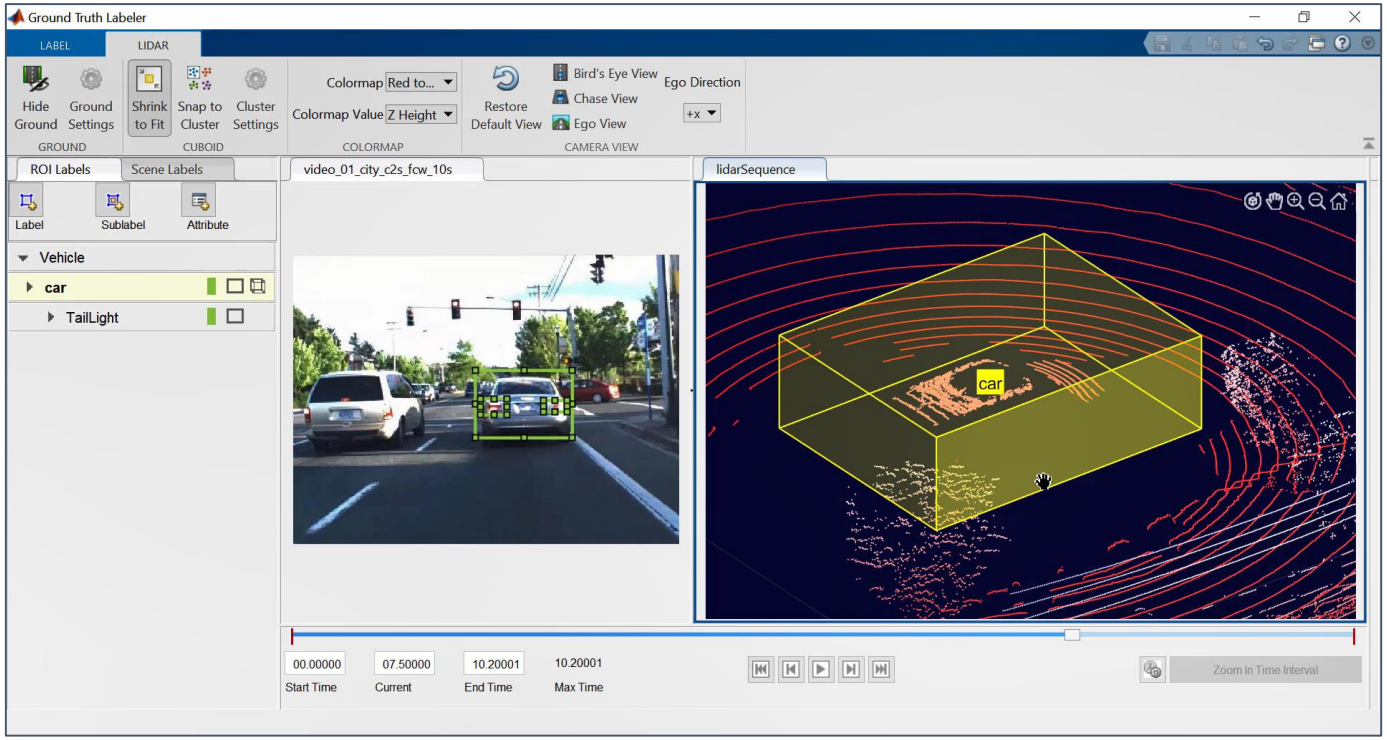
[Get Started with the Ground Truth Labeler]

Automated Driving Toolbox™

Updated R2020a

可视化和标注图像与激光雷达数据

多传感器可视化 → 交互式标注 → 自动标注 → 导出标签



- 交互式标注图像和激光雷达数据

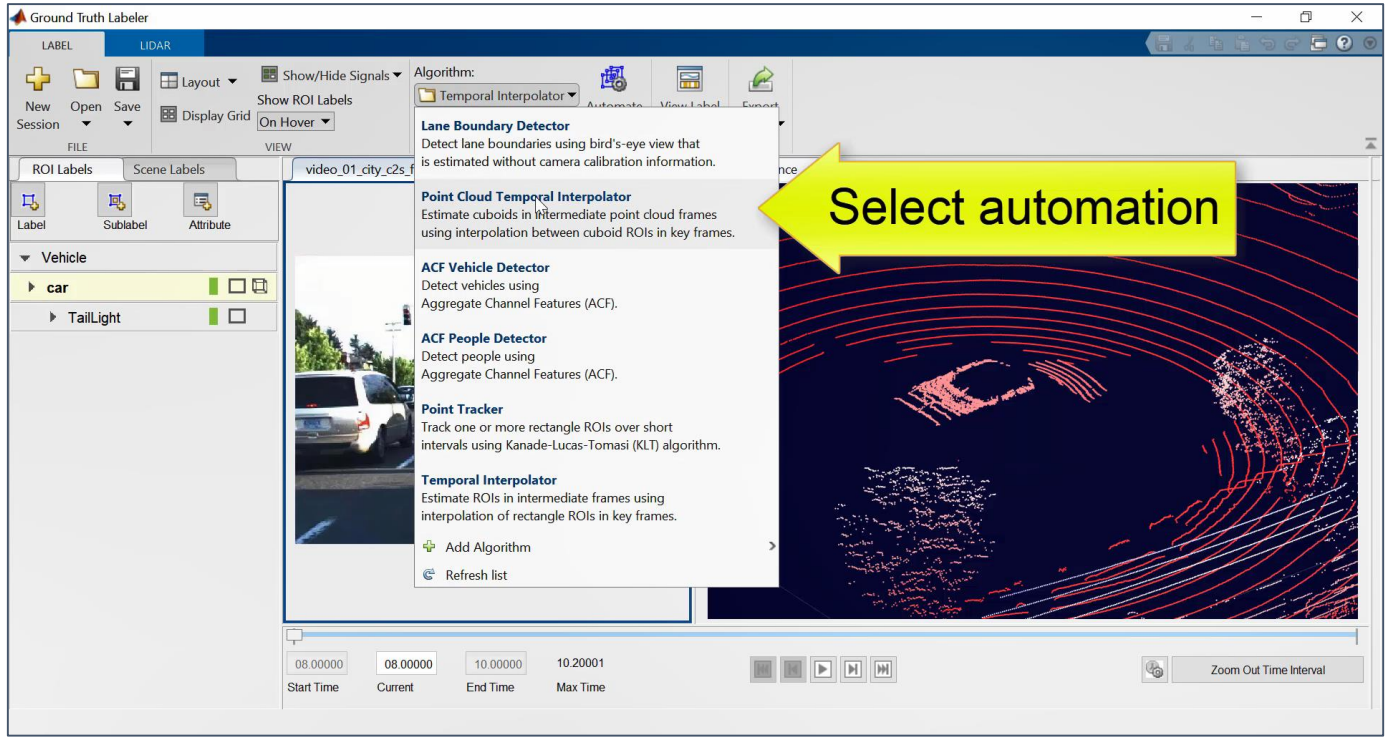
[Get Started with the Ground Truth Labeler]

Automated Driving Toolbox™

Updated R2020a

可视化和标注图像与激光雷达数据

多传感器可视化 → 交互式标注 → 自动标注 → 导出标签



- 使用内置的目标检测与跟踪算法
- 可添加自定义算法，扩展自动化的工作流程

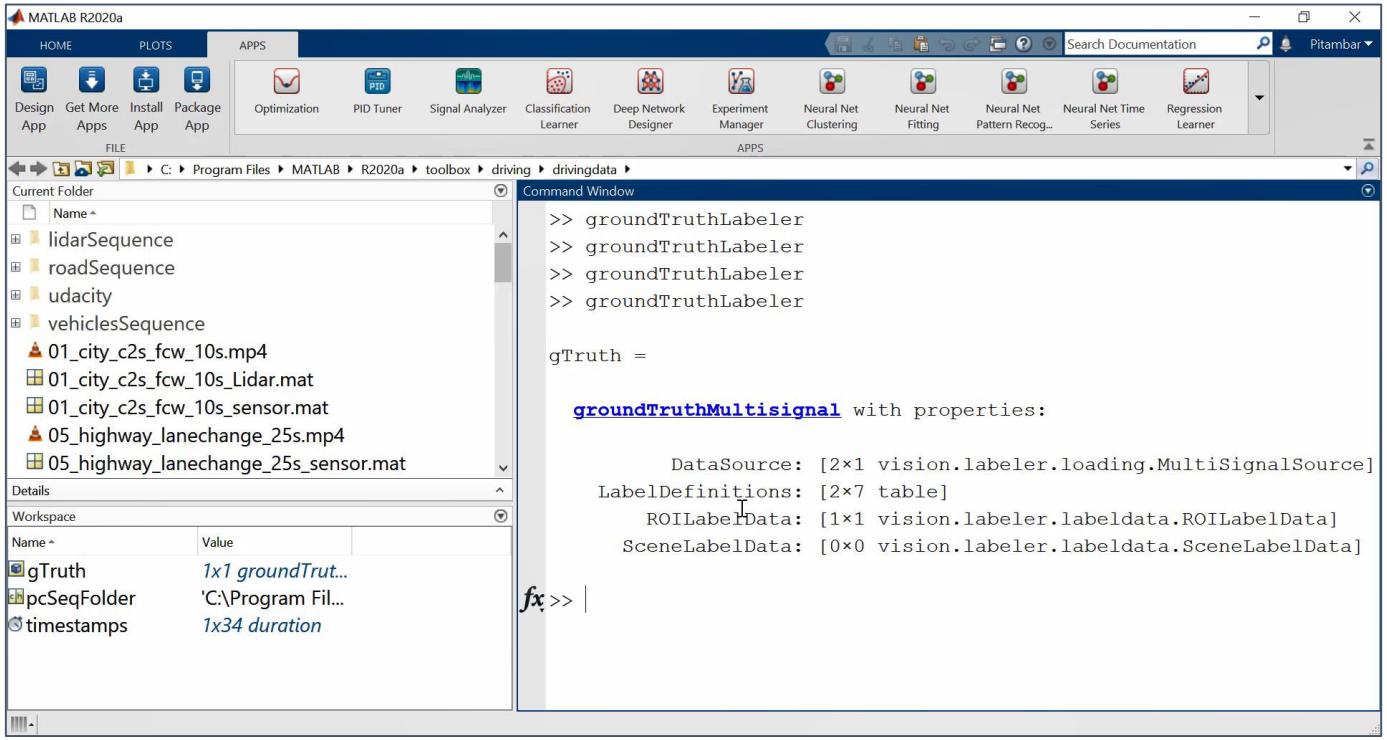
[Get Started with the Ground Truth Labeler]

Automated Driving Toolbox™

Updated R2020a

可视化和标注图像与激光雷达数据

多传感器可视化 → 交互式标注 → 自动标注 → 导出标签



- 导出到工作空间或文件
- 可进一步编辑标签格式，用于与其他工具集成

[Get Started with the Ground Truth Labeler]

Automated Driving Toolbox™

Updated R2020a

分析与合成驾驶场景

分析实车数据的工作流程

数据获取

数据可视化

数据标注

开环仿真

合成虚拟场景的工作流程

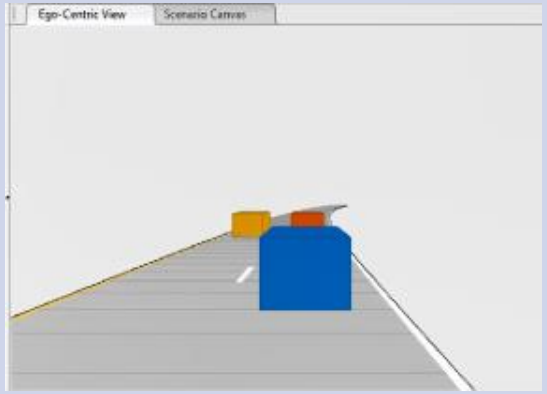
创建环境场景

添加交通参与者

添加传感器

开环或闭环仿真

通过合成虚拟场景测试算法和系统

场景表达	**“方块式”驾驶场景**  	**“虚幻”引擎（UE4）** 
可测试算法	控制，融合，规划	控制，融合，规划，感知
编辑器	MATLAB中的驾驶场景设计器及编程接口	“虚幻”编辑器（Unreal Editor）
传感器模型	基于概率的雷达目标列表 基于概率的视觉目标列表 基于概率的车道线检测列表 激光雷达（点云）	基于概率的雷达目标列表 单目相机（图像、标签、深度） 鱼眼相机（图像） 激光雷达（点云）

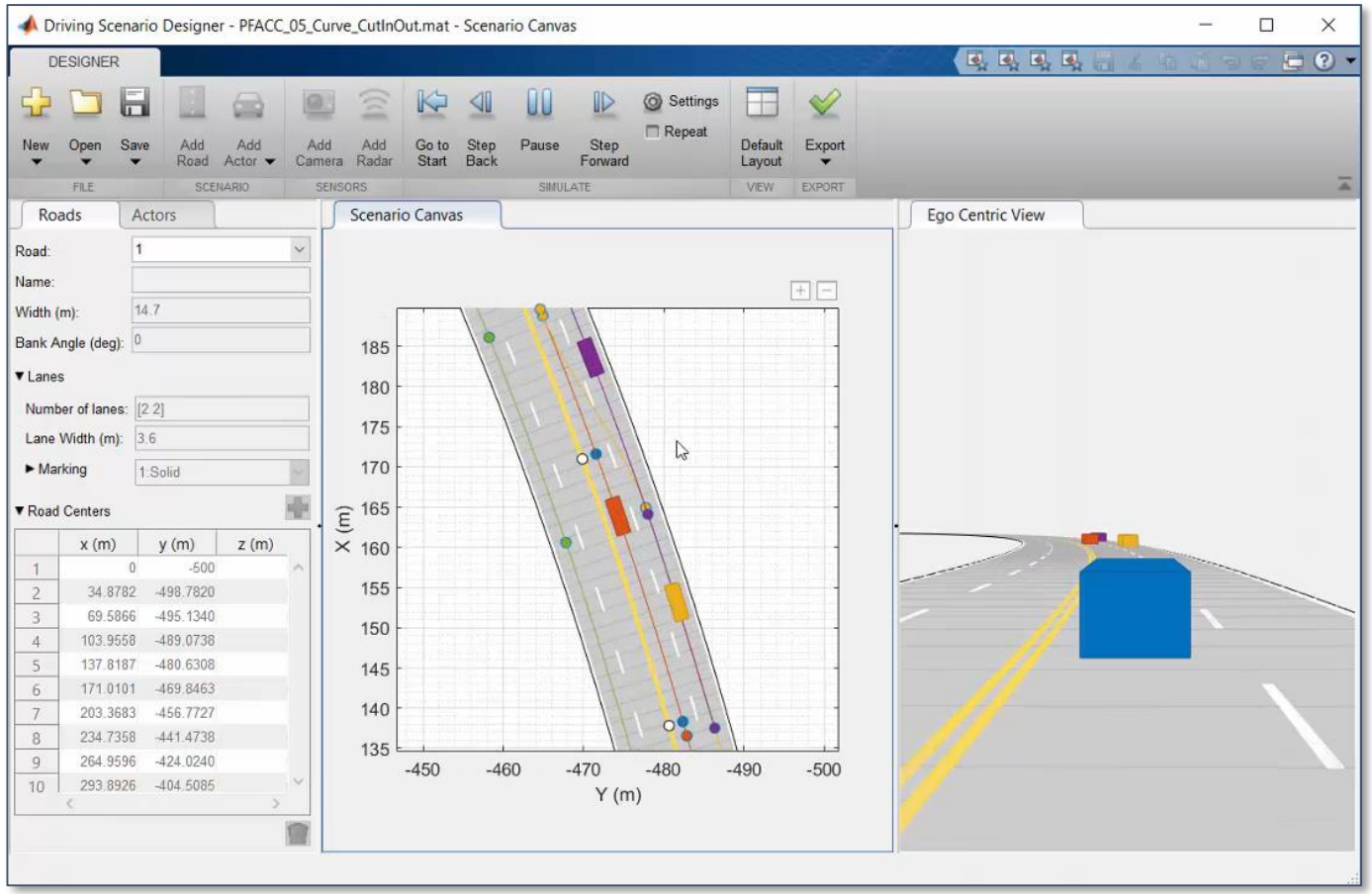
驾驶场景设计器

Driving Scenario Designer

- 创建道路与车道线标记
- 添加车辆与行驶轨迹
- 设置车辆尺寸与雷达截面积(RCS)
- 提供预定义的NCAP场景
- 支持导入OpenDRIVE格式路网文件
- 支持导入HERE高精度实时地图路网

Automated Driving Toolbox™

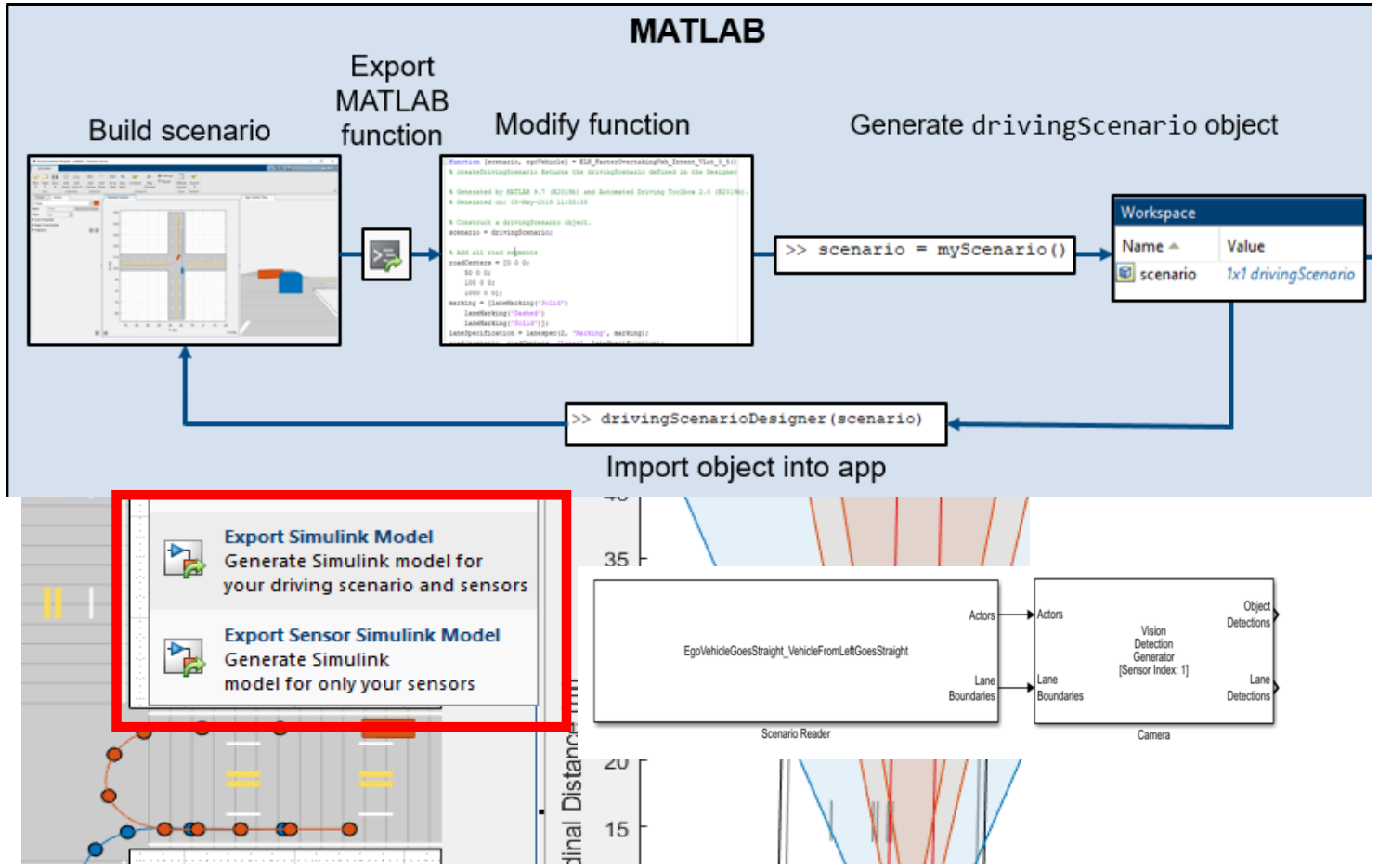
Updated R2020a



驾驶场景设计器的扩展

[Create Driving Scenario Variations Programmatically]

- 导出创建驾驶场景的MATLAB代码并通过编程获得衍生场景
- 导出场景或传感器数据到Simulink并使用该场景测试驾驶算法

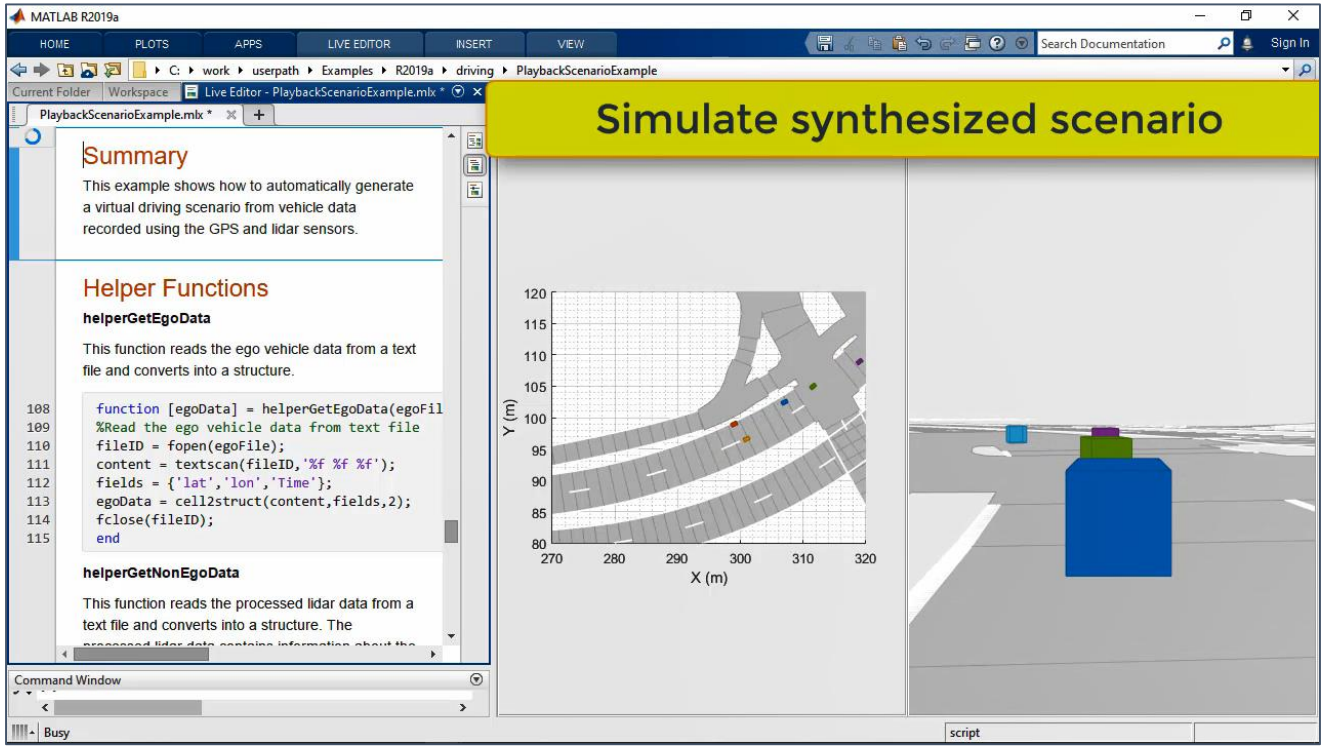


Automated Driving Toolbox™

R2019b

使用实车数据创建自动驾驶测试场景

视频回放 → 导入路网 → 创建主车 → 创建其他车辆 → 仿真场景



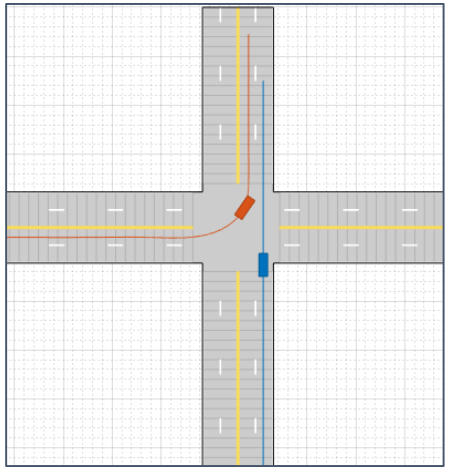
- 导入路网 (来自OpenDRIVE文件)
- 创建主车轨迹 (来自GPS数据)
- 创建其他车辆轨迹 (来自Lidar数据)

[Scenario Generation from Recorded Vehicle Data]

Automated Driving Toolbox™
R2019a

对驾驶场景中的交通参与者建模

场景衍生

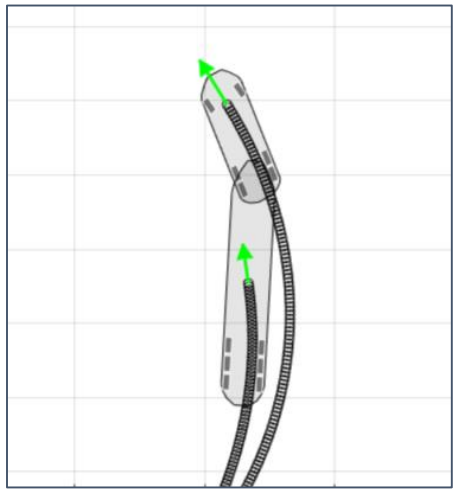


[Create Driving Scenario Variations Programmatically]

Automated Driving Toolbox™

R2019b

车辆动力学

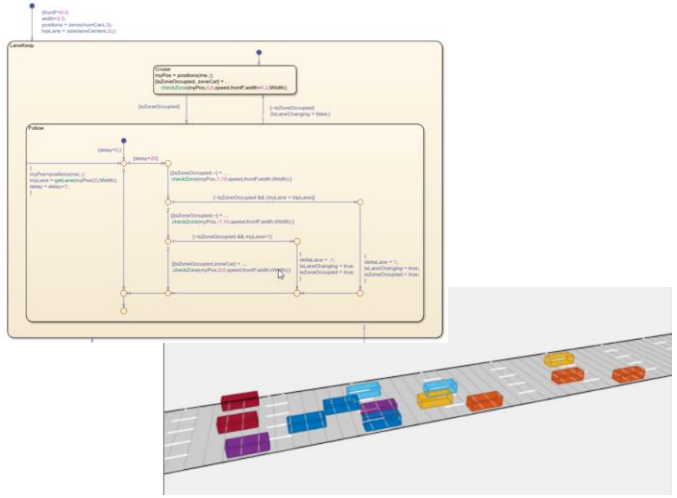


[Three-Axle Tractor Towing a Trailer]

Vehicle Dynamics Blockset™

R2020a

智能车辆



[Automate Control of Intelligent Vehicles by Using Stateflow Charts]

Automated Driving Toolbox™

Stateflow®

R2020a

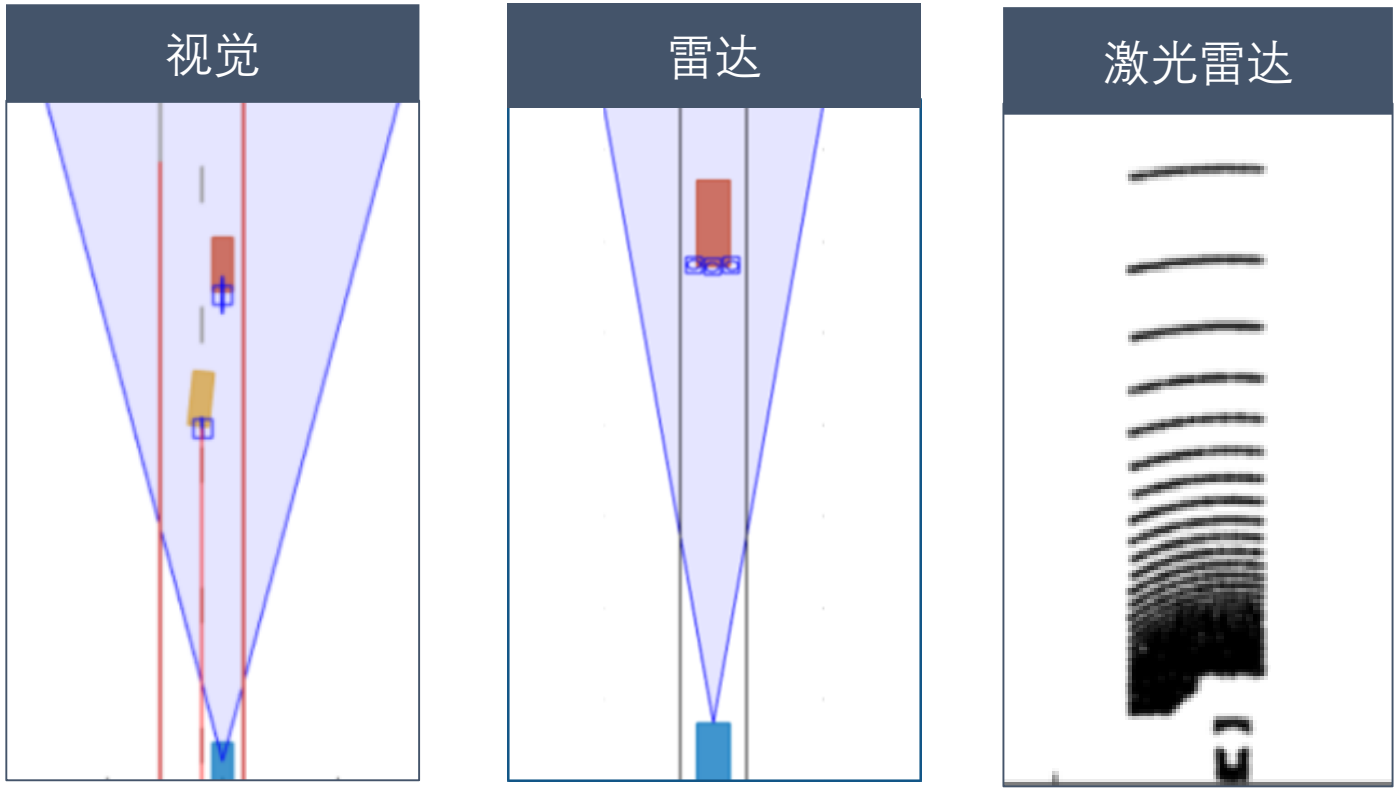
对驾驶场景中的传感器建模

- 视觉目标列表
- 视觉车道线检测
- 雷达目标列表
- 激光雷达点云

Cuboid Driving Scenario Simulation

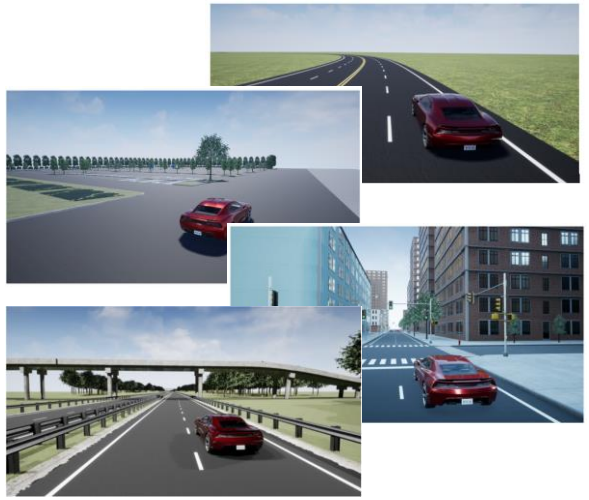
Automated Driving Toolbox™

Updated R2020a



仿真使用“虚幻”引擎创建的3D场景

预定义场景

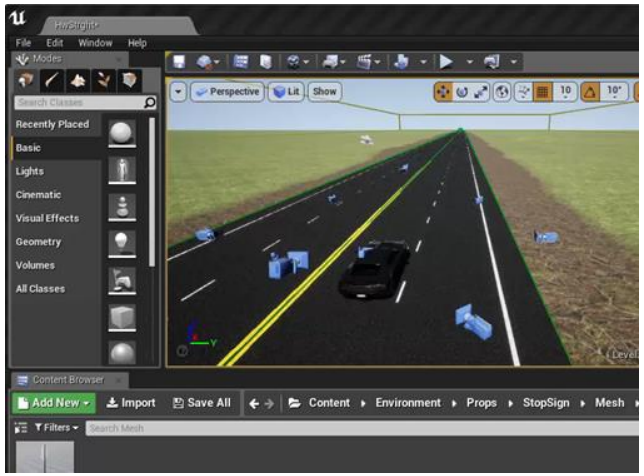


[3D Simulation for Automated Driving]

Automated Driving Toolbox™

R2019b

自定义场景

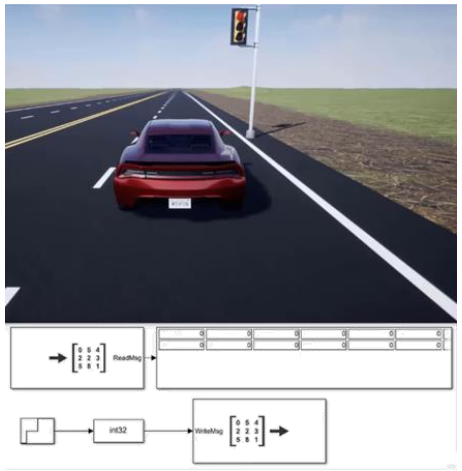


[Customize 3D Scenes for Automated Driving]

Automated Driving Toolbox™

R2020a

自定义消息



[Send and Receive Double-Lane Change Scene Data]

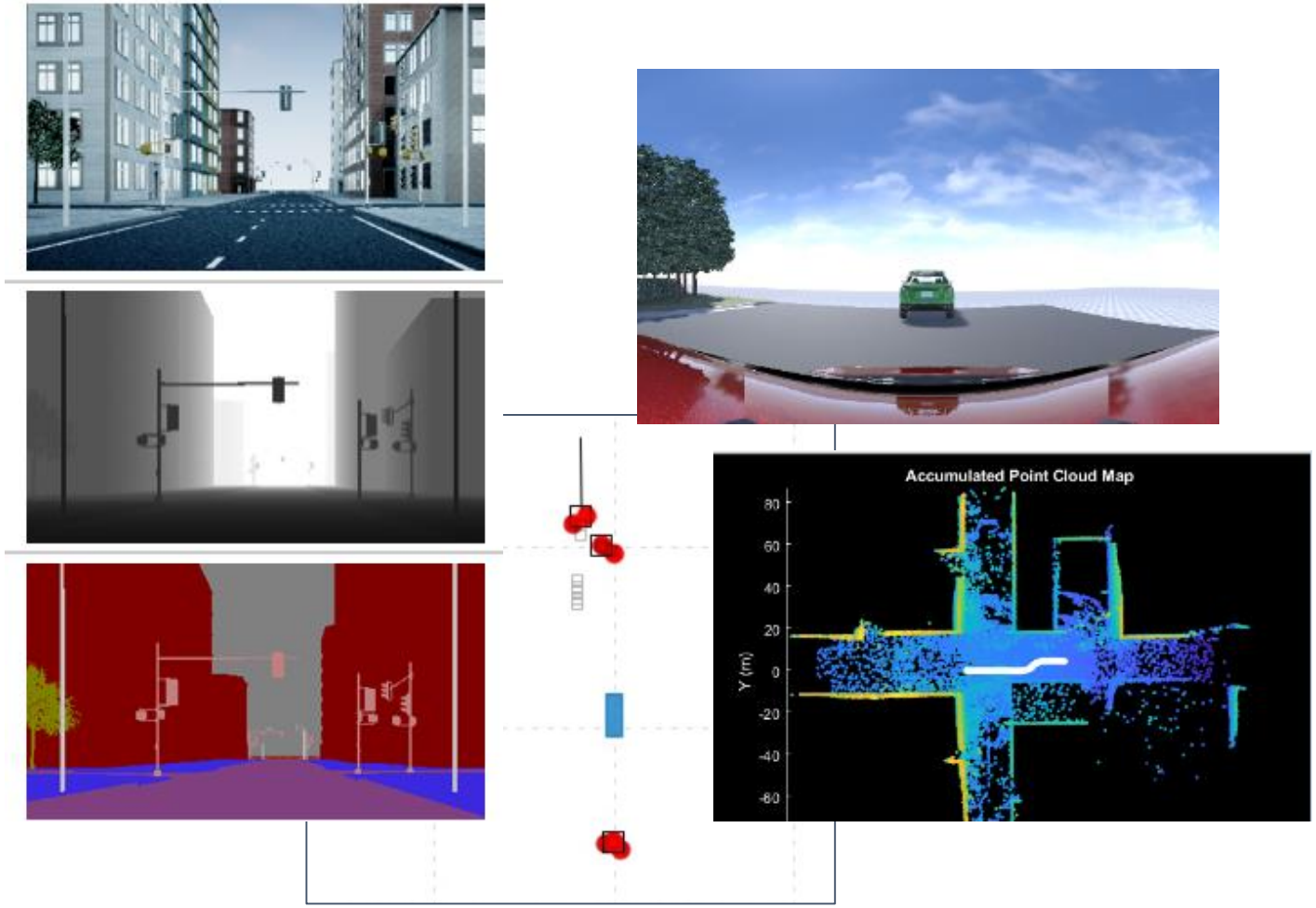
Vehicle Dynamics Toolbox™

R2019b

3D仿真环境中的传感器模型

[3D Simulation for Automated Driving]

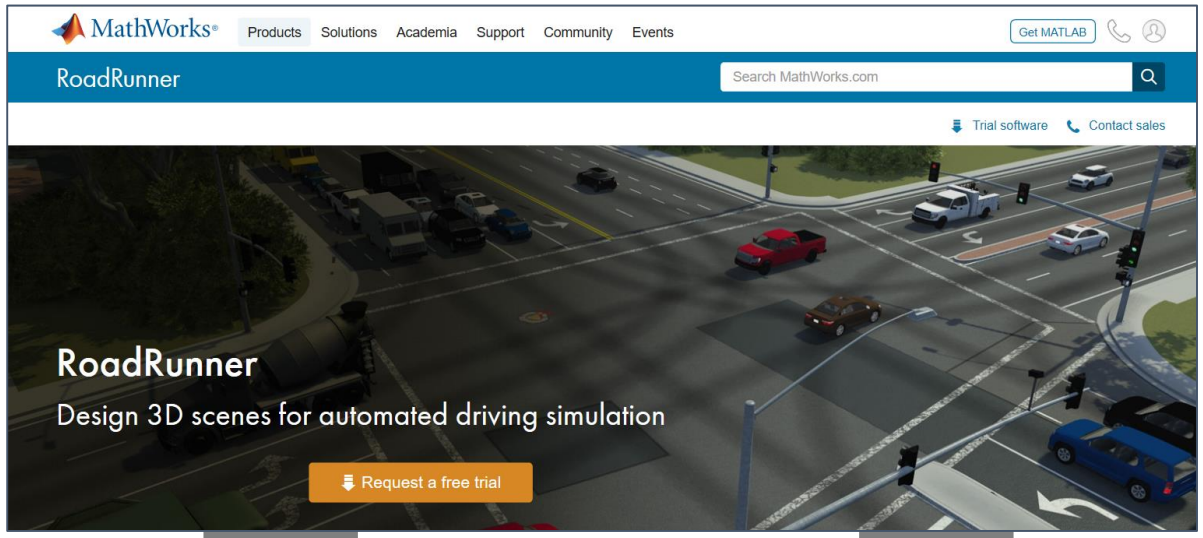
- 单目相机
  - RGB图像
  - 深度图和像素标签
- 鱼眼相机
  - RGB图像
- 激光雷达
  - 点云数据
- 毫米波雷达
  - 目标列表



Automated Driving Toolbox™

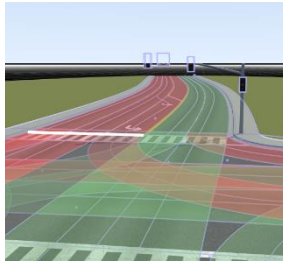
R2019b

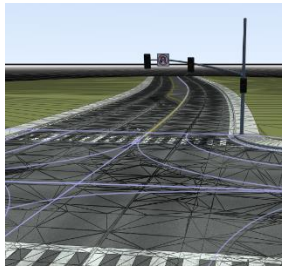
RoadRunner - 为自动驾驶仿真设计3D场景



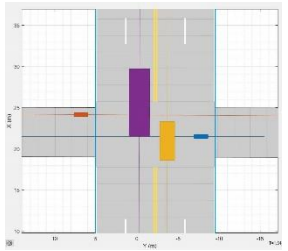
R2020a
Update 1

导出到外部仿真器





MATLAB & Simulink





设计和部署算法

开发规划与控制算法

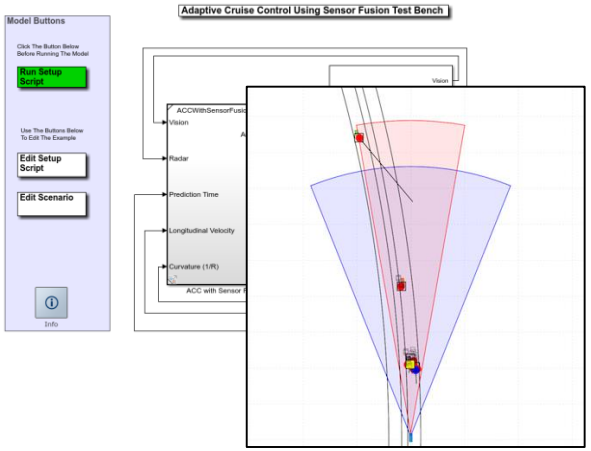
- 运动规划
- 决策逻辑
- 纵向控制
- 横向控制

开发感知算法

- 检测
- 跟踪与融合
- 定位

设计用于ADAS的控制和决策逻辑

Adaptive Cruise Control
(纵向控制)



[Adaptive Cruise Control with Sensor Fusion]

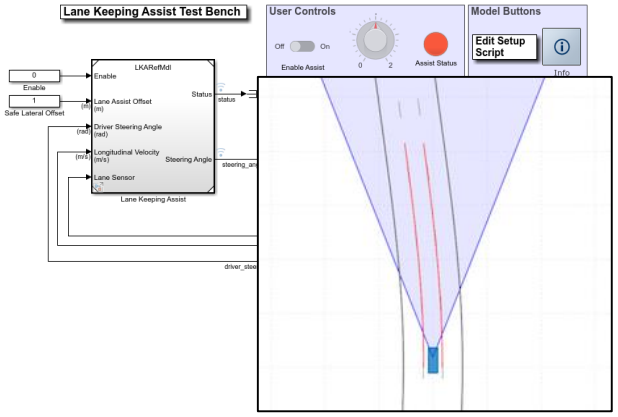
Automated Driving Toolbox™

Model Predictive Control Toolbox™

Embedded Coder®

R2017b

Lane Keep Assist
(横向控制)



[Lane Keeping Assist with Lane Detection]

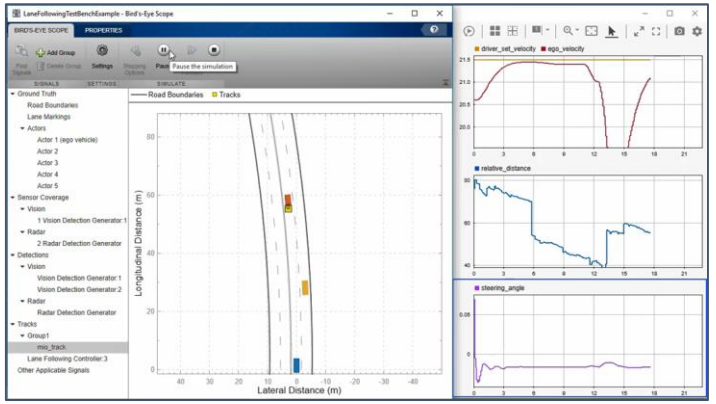
Automated Driving Toolbox™

Model Predictive Control Toolbox™

Embedded Coder®

R2018a

Lane Following
(纵向 + 横向控制)



[Lane Following Control with Sensor Fusion]

Model Predictive Control Toolbox™

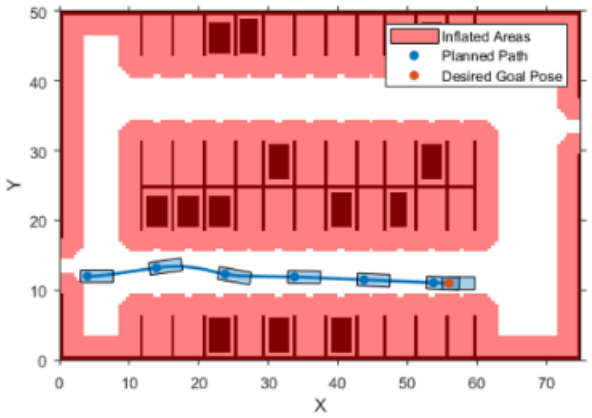
Automated Driving Toolbox™

Embedded Coder®

R2018b

设计用于自动泊车的规划器和控制器

规划器



`pathPlannerRRT`

[Automated Parking Valet]

Automated Driving Toolbox™

R2018a

控制器

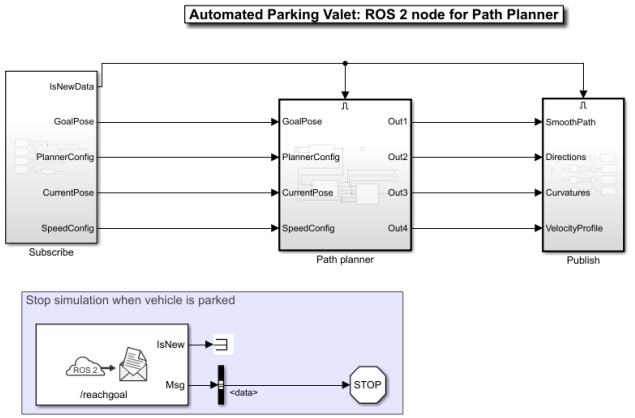


[Automated Parking Valet with Simulink]

Automated Driving Toolbox™

R2018b

生成代码



[Automated Parking Valet with ROS 2 in Simulink]

Automated Driving Toolbox™

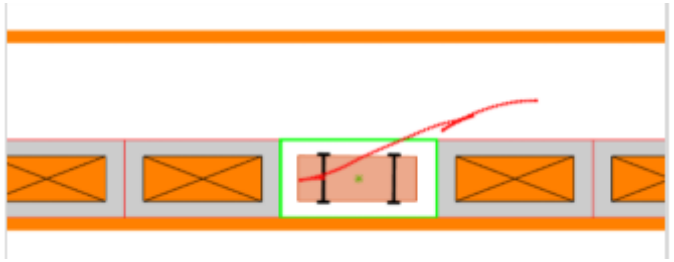
ROS Toolbox™

Embedded Coder®

R2019b

设计含模型预测控制的泊车规划器和控制器

规划器：RRT
控制器：MPC

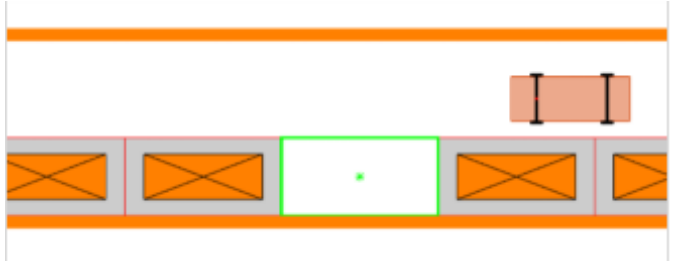


[Parallel Parking using RRT Planner and MPC Tracking Controller]

Automated Driving Toolbox™
Model Predictive Control Toolbox™
Navigation Toolbox™

R2020a

规划器 & 控制器：
NLMPC

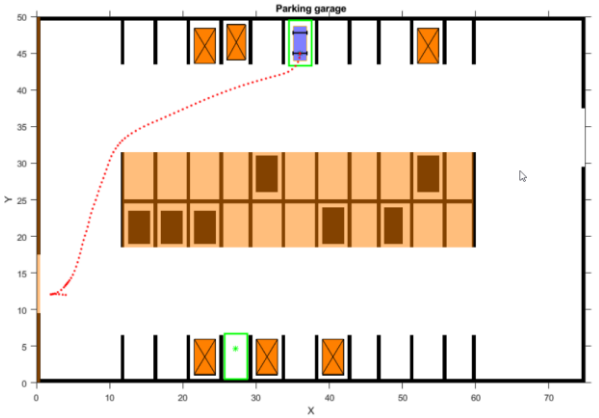


[Parallel Parking using Nonlinear Model Predictive Control]

Automated Driving Toolbox™
Model Predictive Control Toolbox™
Navigation Toolbox™

R2020a

规划器 & 控制器：
NLMPC



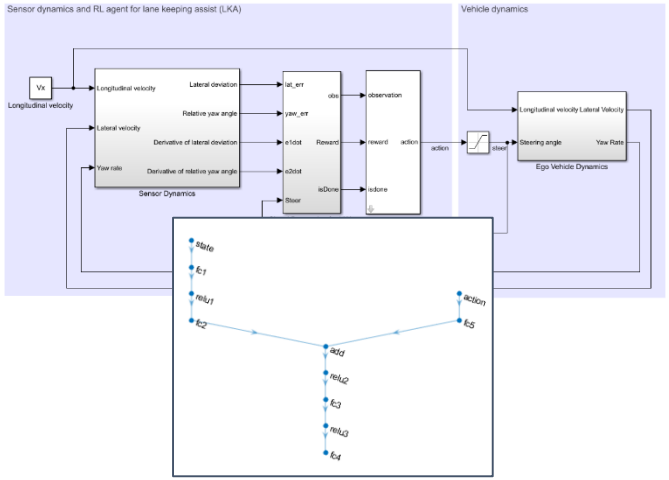
[Parallel Valet using Nonlinear Model Predictive Control]

Automated Driving Toolbox™
Model Predictive Control Toolbox™
Navigation Toolbox™

R2020a

设计基于强化学习的控制器

训练全新网络

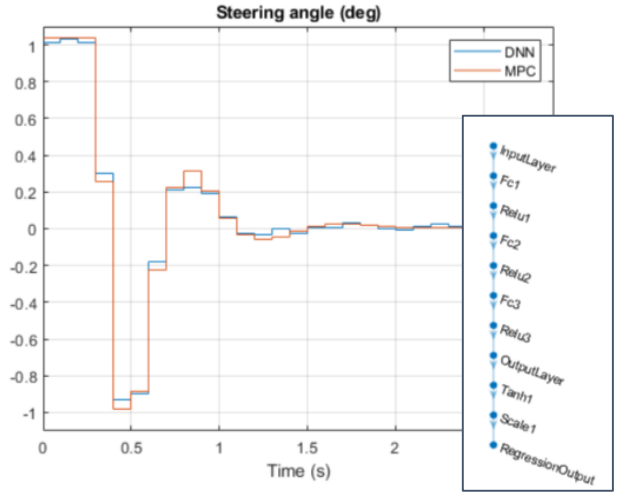


[Train DQN Agent for Lane Keeping Assist]

Reinforcement Learning Toolbox™

R2019a

模仿MPC控制



[Imitate MPC Controller for Lane Keep Assist]

Reinforcement Learning Toolbox™

Model Predictive Control Toolbox™

R2020a

使用预训练的网络



[Train DDPG Agent with Pretrained Actor Network]

Reinforcement Learning Toolbox™

R2020a

设计和部署算法

开发规划与控制算法

- 运动规划
- 决策逻辑
- 纵向控制
- 横向控制

开发感知算法

- 检测
- 跟踪与融合
- 定位

基于深度学习的检测器与分类器

SSD (single shot detector)



[Object Detection Using SSD Deep Learning]

Computer Vision Toolbox™

Deep Learning Toolbox™

R2020a

YOLO v3 (you only look once)



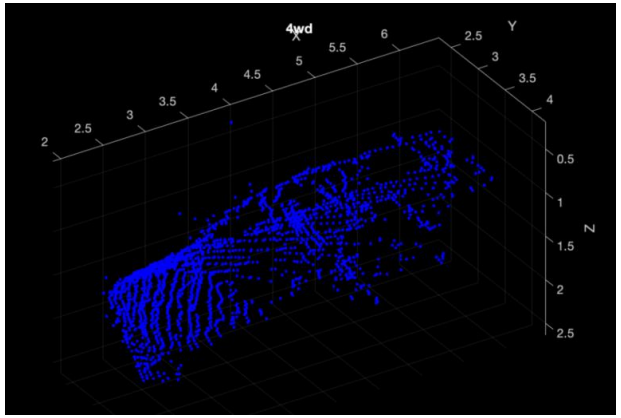
[Object Detection Using YOLO v3 Deep Learning]

Computer Vision Toolbox™

Deep Learning Toolbox™

R2020a

PointNet



[Point Cloud Classification Using PointNet Deep Learning]

Computer Vision Toolbox™

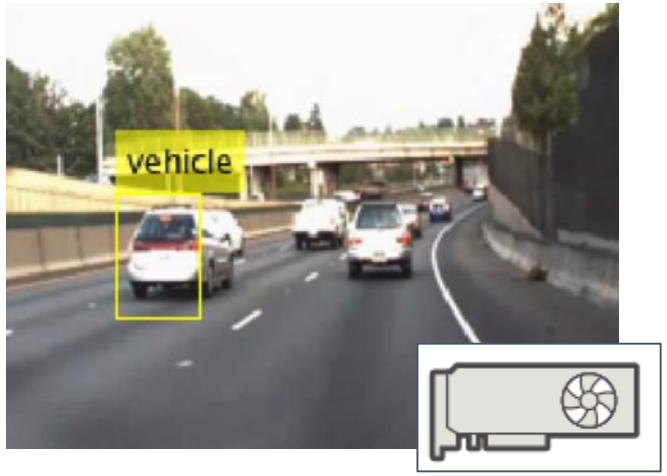
Deep Learning Toolbox™

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部署深度学习网络

NVIDIA GPU



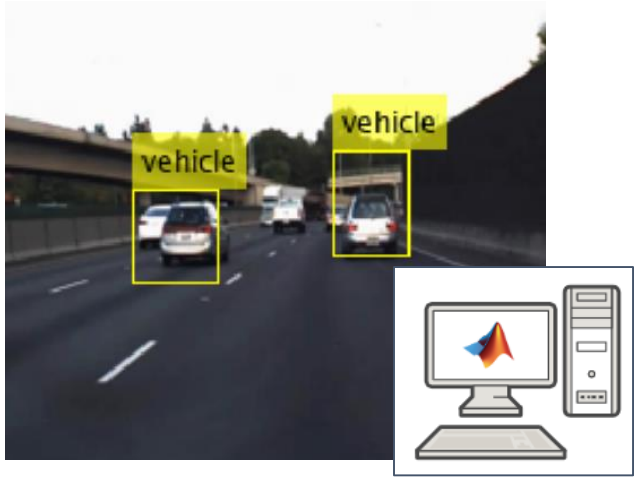
[Code Generation for Object Detection by Using Single Shot Multibox Detector](#)

Deep Learning Toolbox™

GPU Coder™

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Intel MKL-DNN



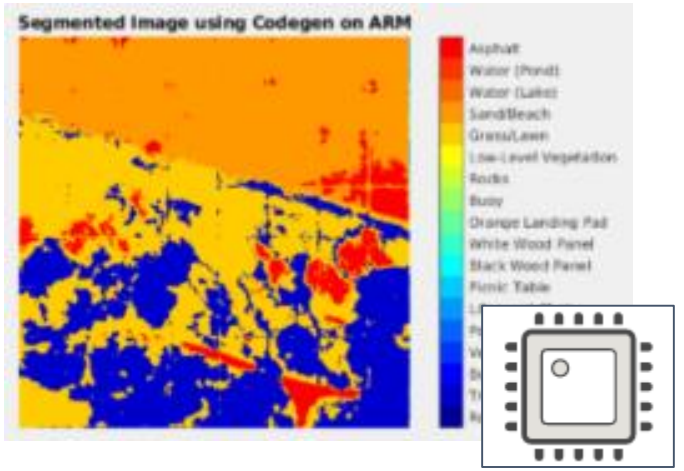
[Generate C++ Code for Object Detection Using YOLO v2 and Intel MKL-DNN](#)

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ARM



[Code Generation for Semantic Segmentation Application on ARM Neon](#)

Deep Learning Toolbox™

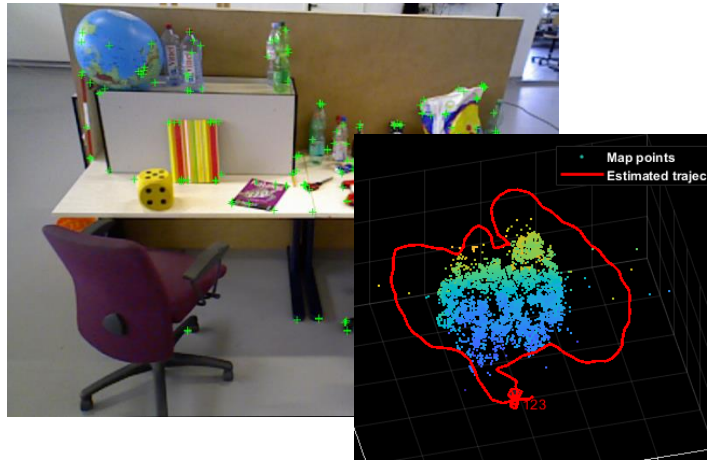
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设计SLAM系统

单目相机



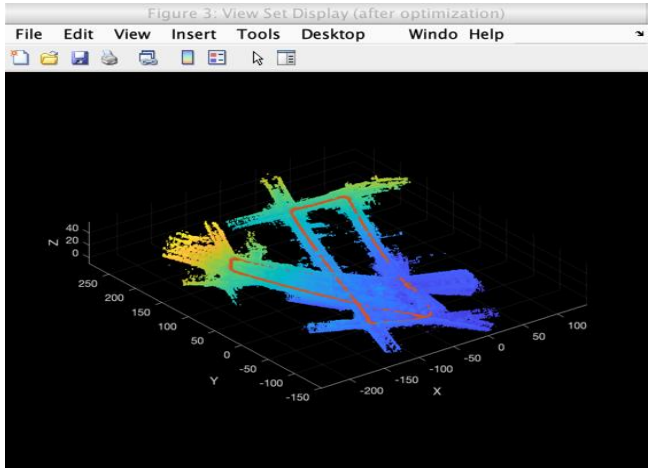
[Monocular Visual Simultaneous Localization and Mapping]

Automated Driving Toolbox™

Computer Vision Toolbox™

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激光雷达（真实数据）



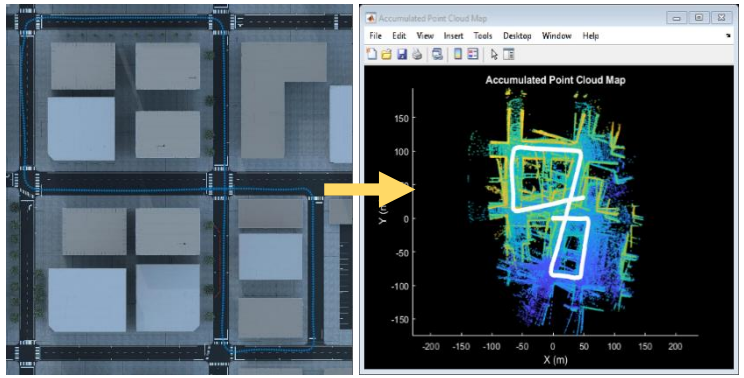
[Build a Map from Lidar Data Using SLAM]

Automated Driving Toolbox™

Computer Vision Toolbox™

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激光雷达（合成数据）



[Design Lidar SLAM Algorithm using 3D Simulation Environment]

Automated Driving Toolbox™

Computer Vision Toolbox™

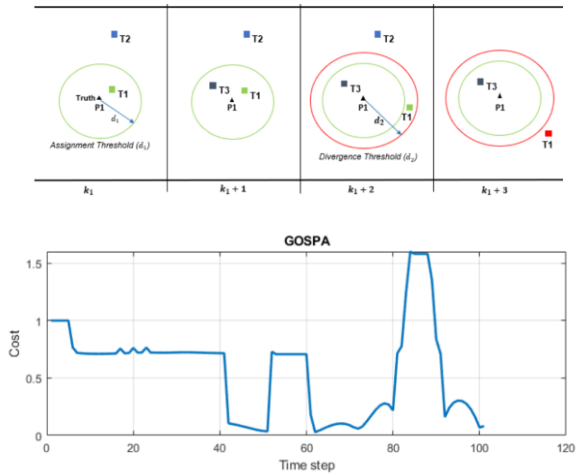
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MathWorks AUTOMOTIVE CONFERENCE

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目标跟踪与传感器融合

评估

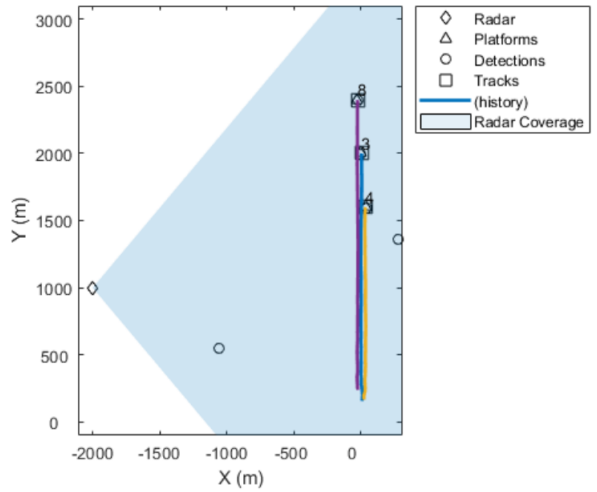


[Introduction to Tracking Metrics]

Sensor Fusion and Tracking Toolbox™

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调参

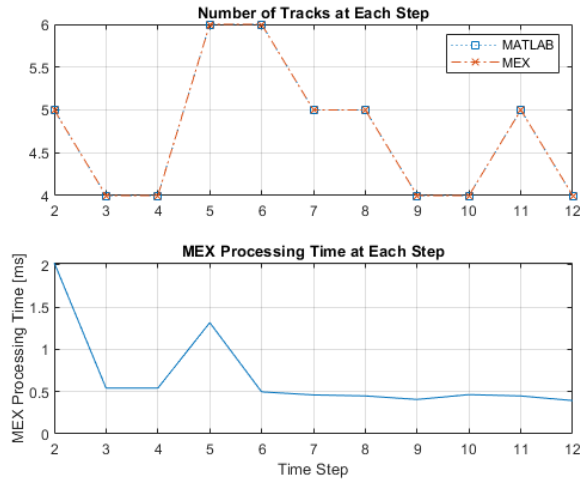


[Tuning a Multi-Object Tracker]

Sensor Fusion and Tracking Toolbox™

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生成代码



[Generate C Code for a Tracker]

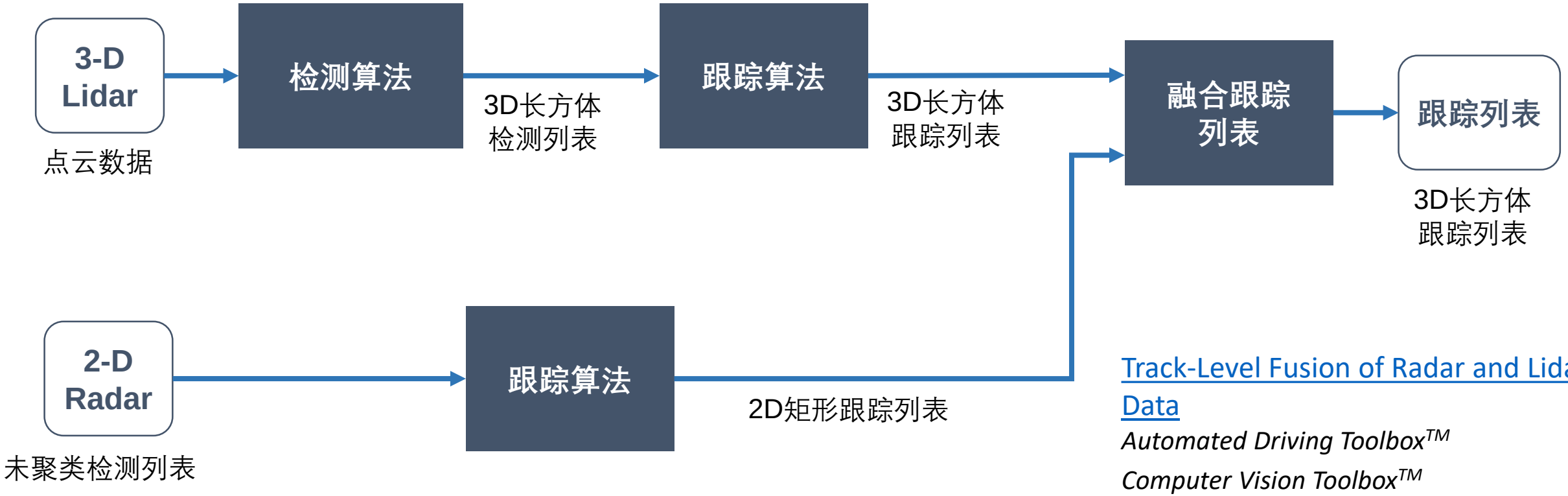
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融合激光雷达和毫米波雷达



[Track-Level Fusion of Radar and Lidar Data]

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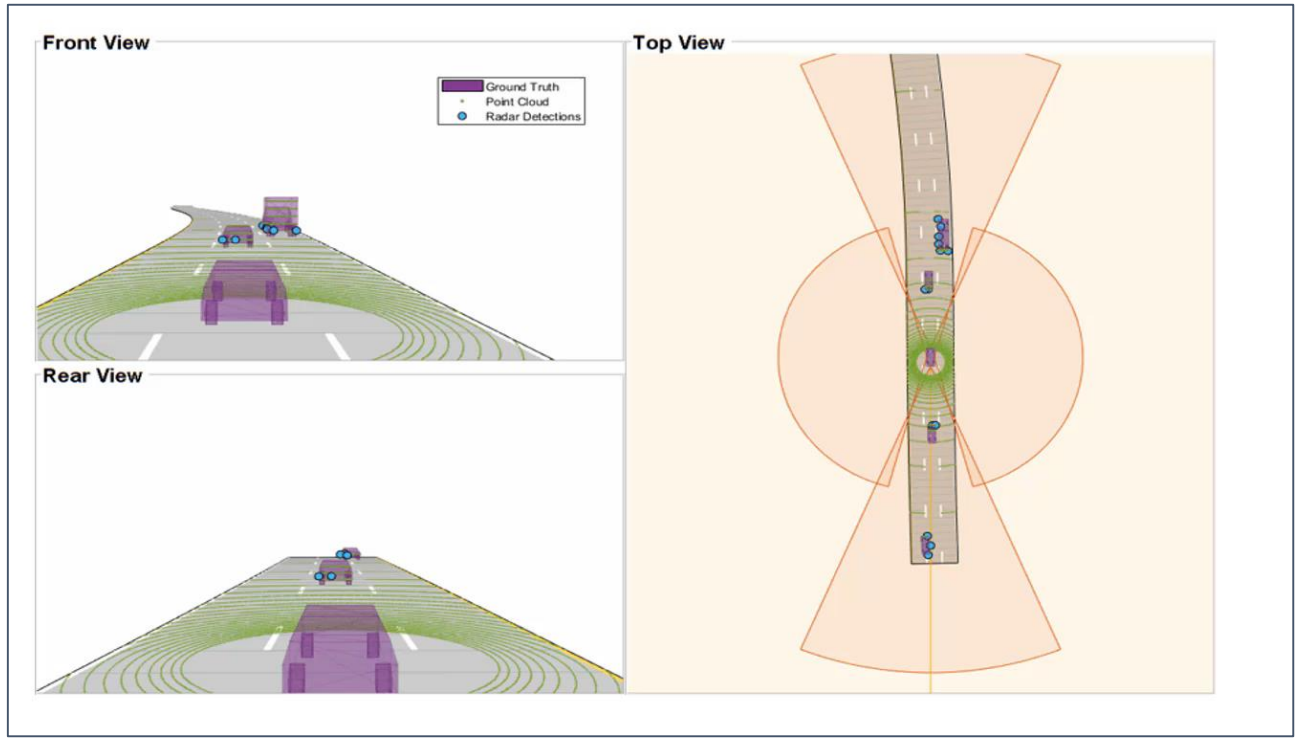
Computer Vision Toolbox™

Sensor Fusion and Tracking Toolbox™

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融合激光雷达和毫米波雷达

编辑场景 → 检测器 (激光雷达) → 跟踪器 (激光雷达) → 跟踪器 (毫米波雷达) → 融合 → 评估指标



- 创建场景
- 添加参与者
- 添加激光雷达点云传感器
- 添加毫米波雷达传感器

Track-Level Fusion of Radar and Lidar Data

Automated Driving Toolbox™

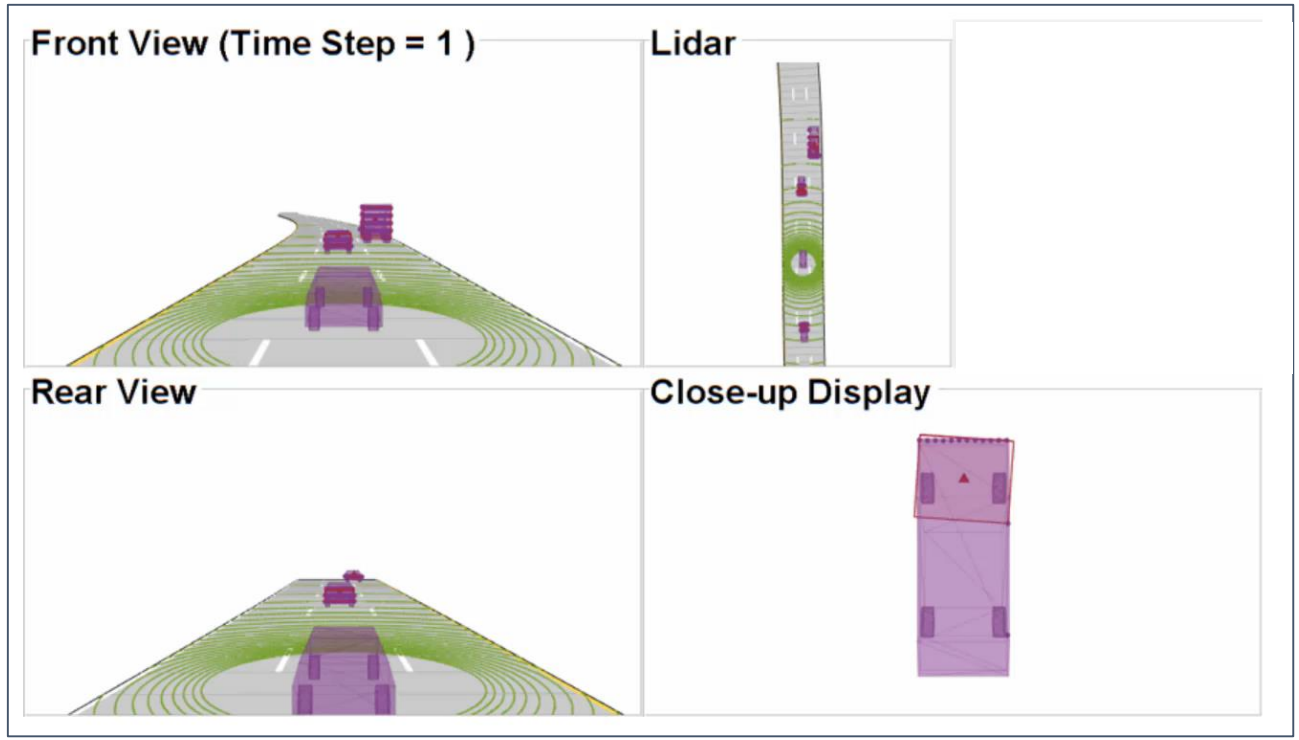
Computer Vision Toolbox™

Sensor Fusion and Tracking Toolbox™

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融合激光雷达和毫米波雷达

编辑场景 → 检测器 (激光雷达) → 跟踪器 (激光雷达) → 跟踪器 (毫米波雷达) → 融合 → 评估指标



- 移除地面
- 分割与聚类
- 添加长方体选框

[Track-Level Fusion of Radar and Lidar Data]

Automated Driving Toolbox™

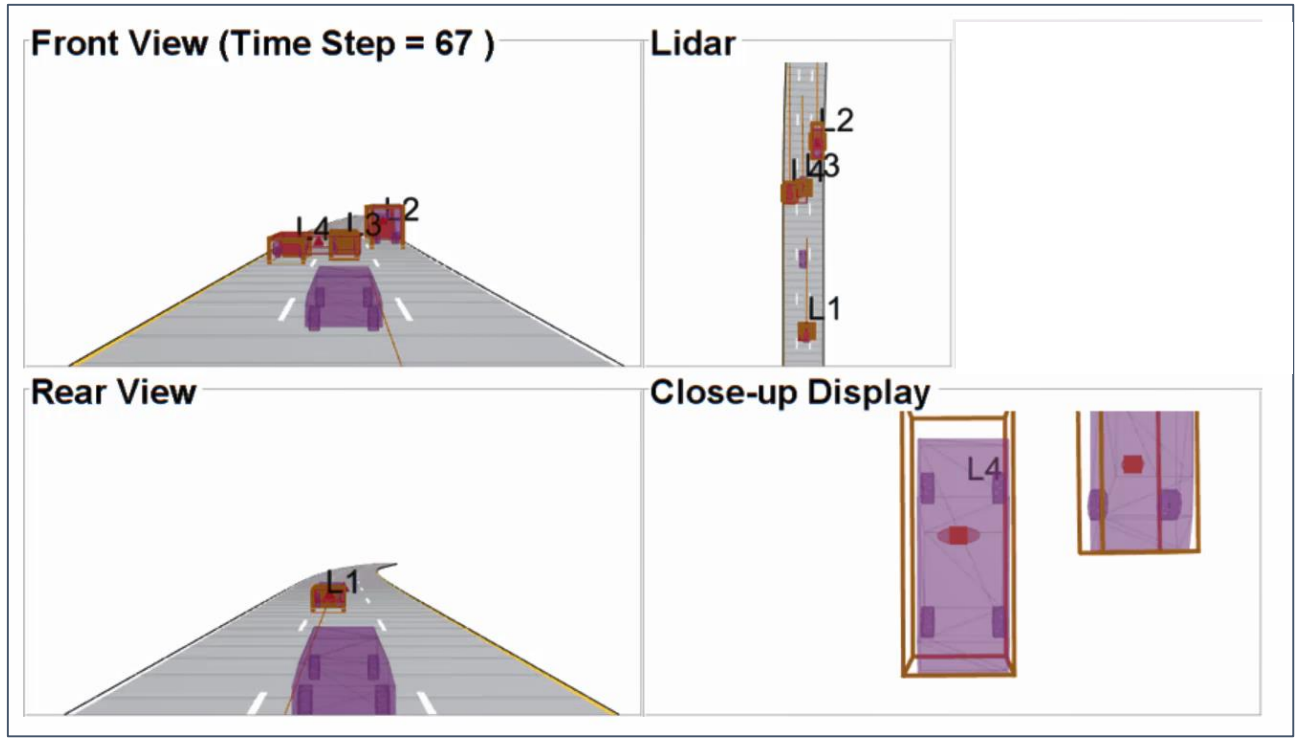
Computer Vision Toolbox™

Sensor Fusion and Tracking Toolbox™

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融合激光雷达和毫米波雷达

编辑场景 → 检测器 (激光雷达) → **跟踪器 (激光雷达)** → 跟踪器 (毫米波雷达) → 融合 → 评估指标



- 设计联合概率数据关联 (JPDA) 多目标跟踪器
- 使用交互式多模型-无迹卡尔曼滤波器 (IMM-UKF)

Track-Level Fusion of Radar and Lidar Data

Automated Driving Toolbox™

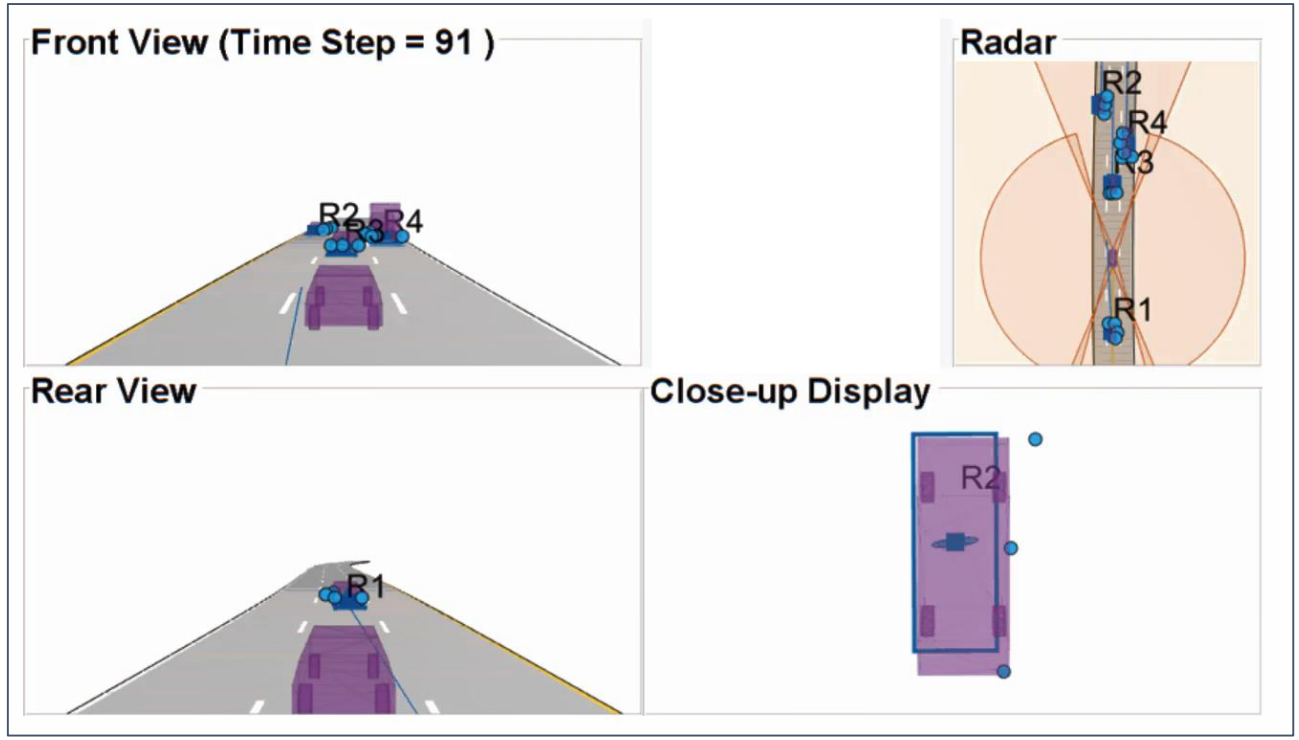
Computer Vision Toolbox™

Sensor Fusion and Tracking Toolbox™

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融合激光雷达和毫米波雷达

编辑场景 → 检测器 (激光雷达) → 跟踪器 (激光雷达) → 跟踪器 (毫米波雷达) → 融合 → 评估指标



- 设计矩形扩展目标跟踪器
- 采用高斯混合-概率假设密度 (GM-PHD) 滤波算法

[Track-Level Fusion of Radar and Lidar Data](#)

Automated Driving Toolbox™

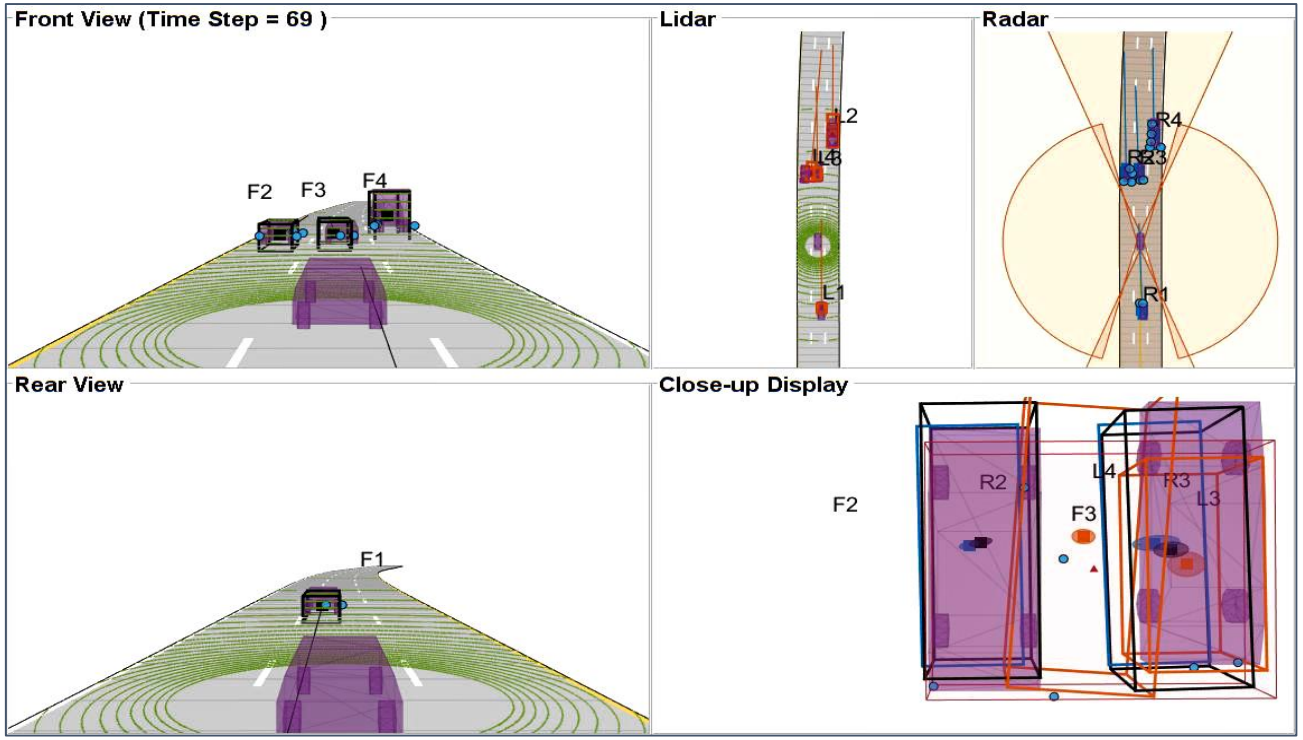
Computer Vision Toolbox™

Sensor Fusion and Tracking Toolbox™

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融合激光雷达和毫米波雷达

编辑场景 → 检测器 (激光雷达) → 跟踪器 (激光雷达) → 跟踪器 (毫米波雷达) → **融合** → 评估指标



- 融合两个跟踪列表
- 协方差交叉融合算法
- 可视化

[Track-Level Fusion of Radar and Lidar Data](#)

Automated Driving Toolbox™

Computer Vision Toolbox™

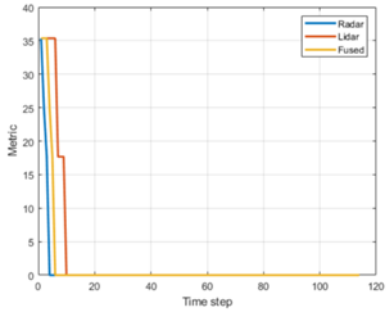
Sensor Fusion and Tracking Toolbox™

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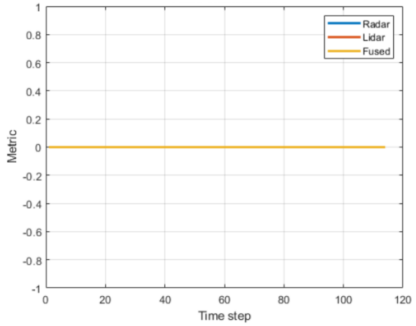
融合激光雷达和毫米波雷达

编辑场景 → 检测器 (激光雷达) → 跟踪器 (激光雷达) → 跟踪器 (毫米波雷达) → 融合 → 评估指标

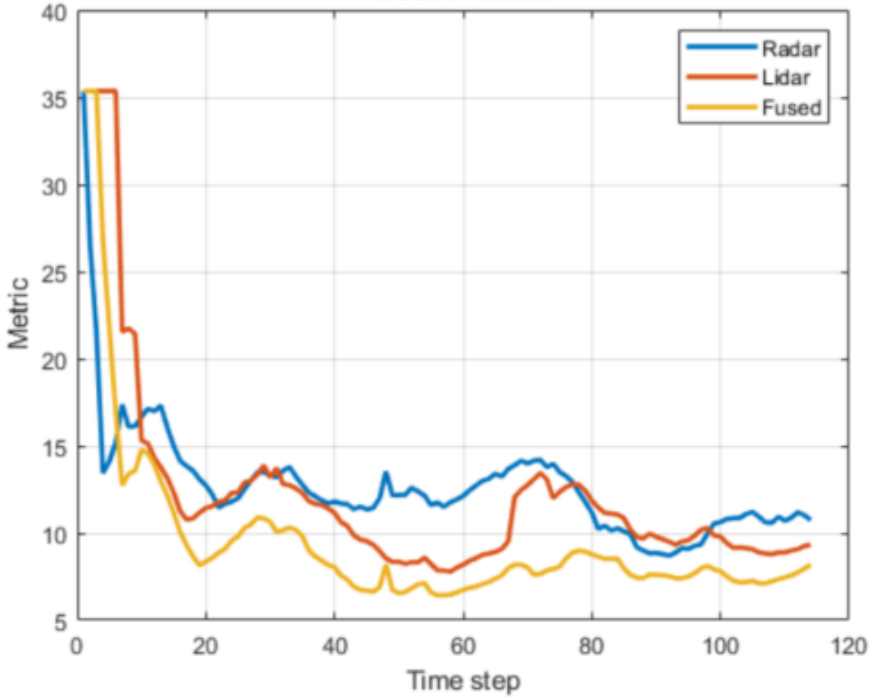
Missed Targets



False Tracks



GOSPA



- 统计丢失目标的数量
- 统计错误跟踪的数量
- 评估广义最优次模式分配 (GOSPA) 指标

[Track-Level Fusion of Radar and Lidar Data]

Automated Driving Toolbox™

Computer Vision Toolbox™

Sensor Fusion and Tracking Toolbox™

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系统的集成与测试

系统集成

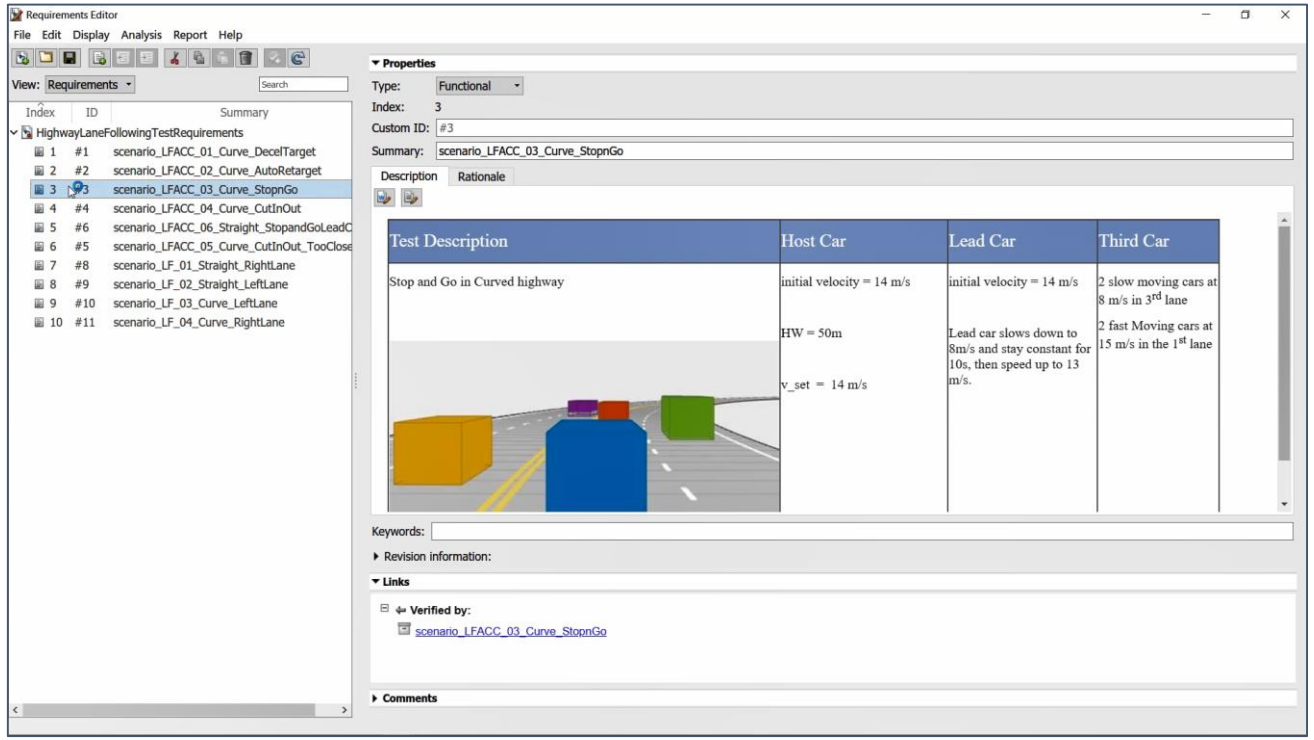
- MATLAB & Simulink
- C / C++ GPU
- CAN ROS
- FMI FMU
- Python
- …

系统测试

- 需求管理
- 自动化
- 功能评估
- 代码评估

自动测试高速公路车道跟随系统

链接需求 → 自动测试 → 评估功能 → 集成代码 → 评估代码



- 编写和关联测试需求和场景

[Automate Testing for Highway Lane Following](#)

Automated Driving Toolbox™

Model Predictive Control Toolbox™

Simulink Test™

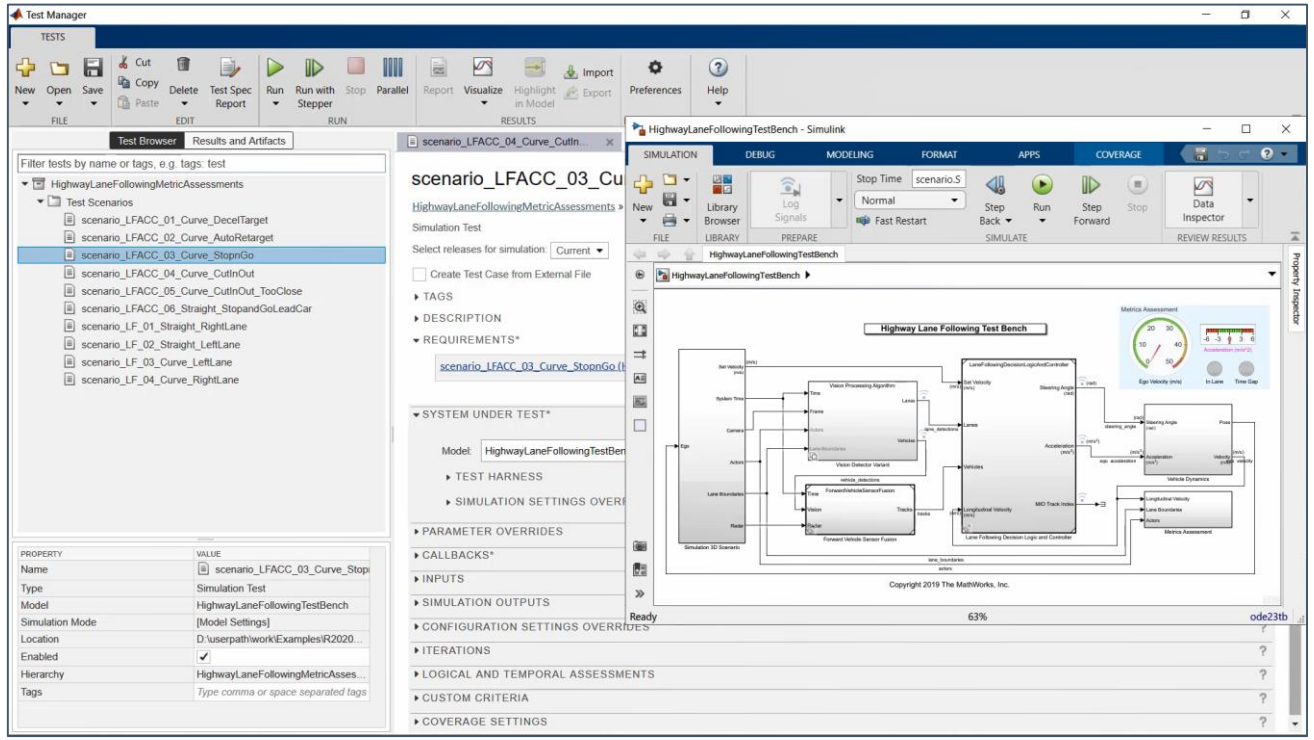
Simulink Requirements™

Simulink Coverage™

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自动测试高速公路车道跟随系统

链接需求 | 自动测试 | 评估功能 | 集成代码 | 评估代码



- 自动运行测试和生成报告
- 采用并行方式同时运行多个仿真

[Automate Testing for Highway Lane Following]

Automated Driving Toolbox™

Model Predictive Control Toolbox™

Simulink Test™

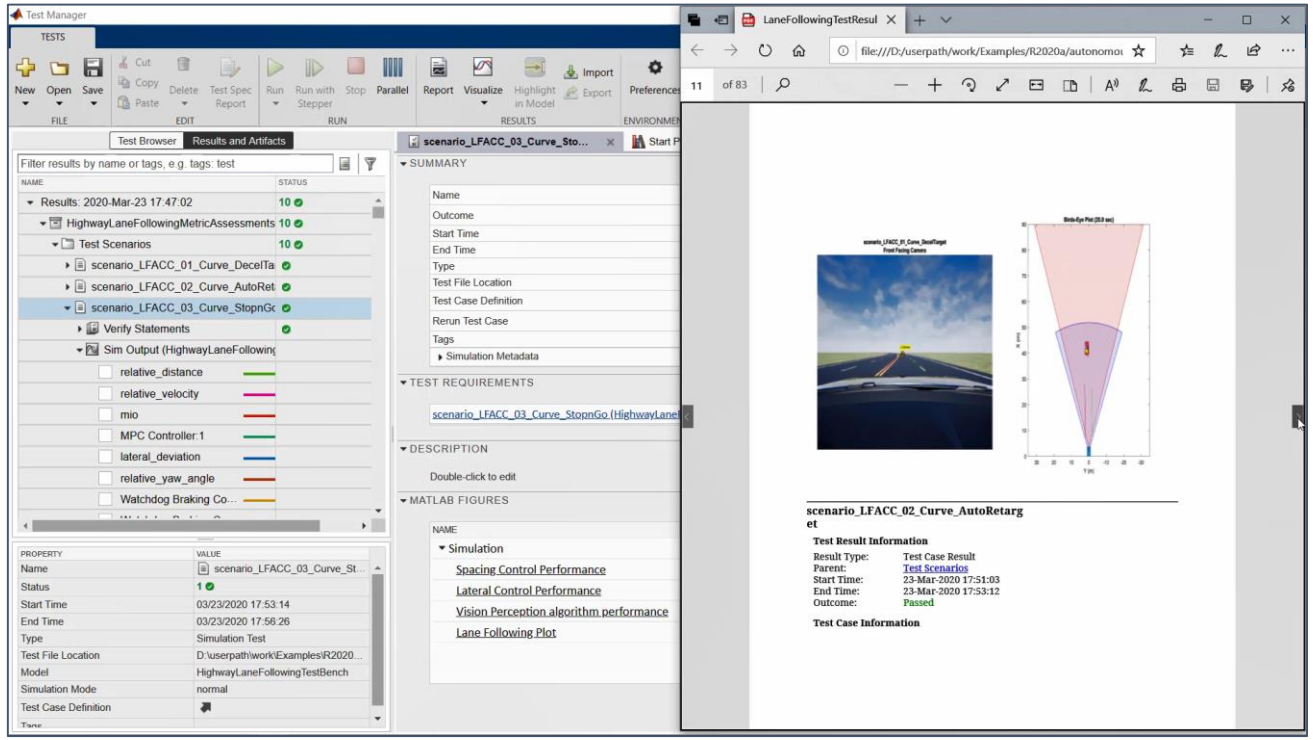
Simulink Requirements™

Simulink Coverage™

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自动测试高速公路车道跟随系统

链接需求 → 自动测试 → 评估功能 → 集成代码 → 评估代码



- 评估纵向和横向控制性能
- 评估车道线检测性能

[Automate Testing for Highway Lane Following]

Automated Driving Toolbox™

Model Predictive Control Toolbox™

Simulink Test™

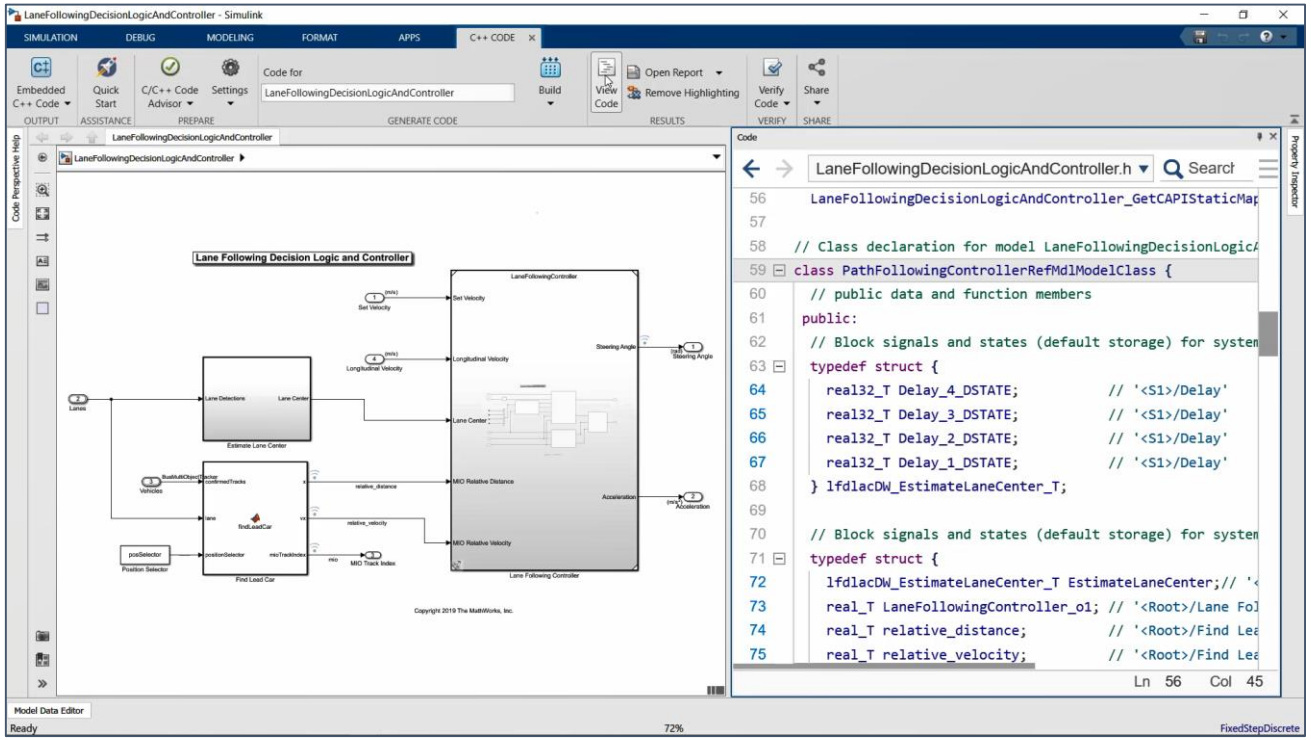
Simulink Requirements™

Simulink Coverage™

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自动测试高速公路车道跟随系统

链接需求 → 自动测试 → 评估功能 → 集成代码 → 评估代码



- 通过软件在环 (SIL) 方式仿真
- 也可用于测试人工编写的代码

[Automate Testing for Highway Lane Following](#)

Automated Driving Toolbox™

Model Predictive Control Toolbox™

Simulink Test™

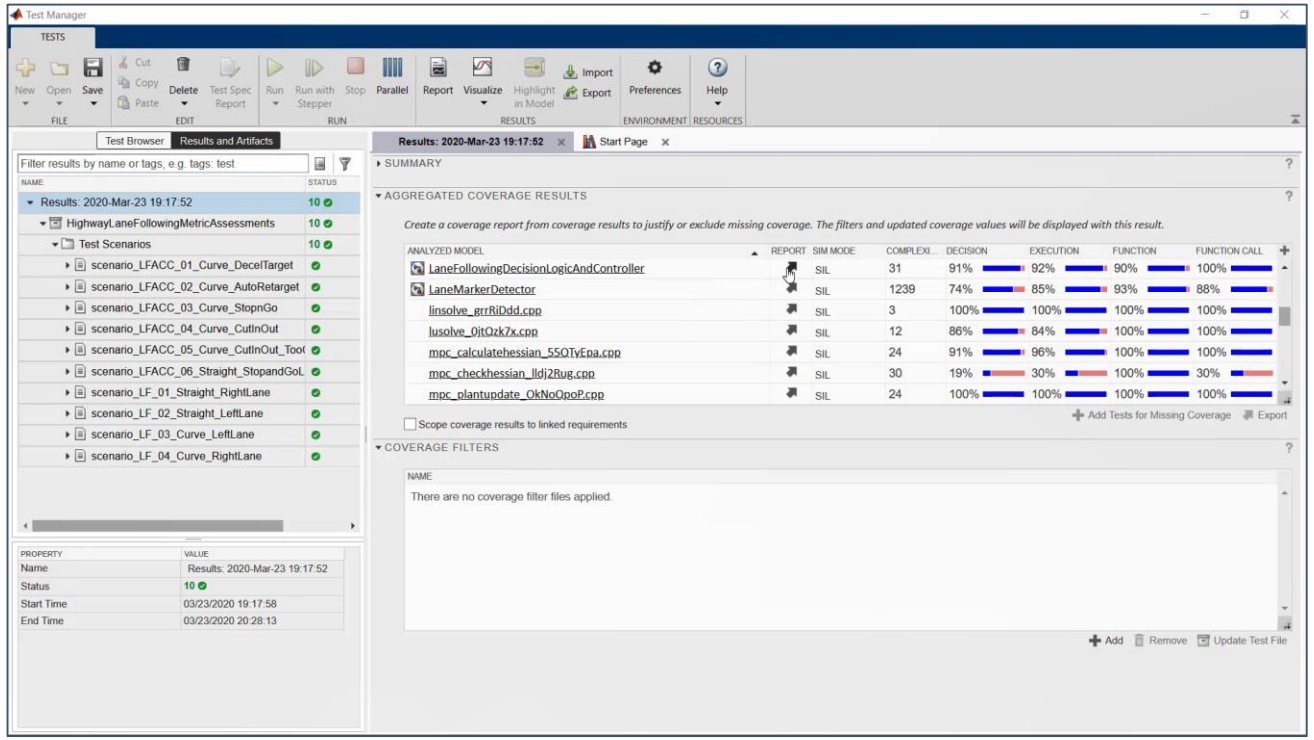
Simulink Requirements™

Simulink Coverage™

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自动测试高速公路车道跟随系统

链接需求 → 自动测试 → 评估功能 → 集成代码 → 评估代码



- 评估代码覆盖率

[Automate Testing for Highway Lane Following](#)

Automated Driving Toolbox™

Model Predictive Control Toolbox™

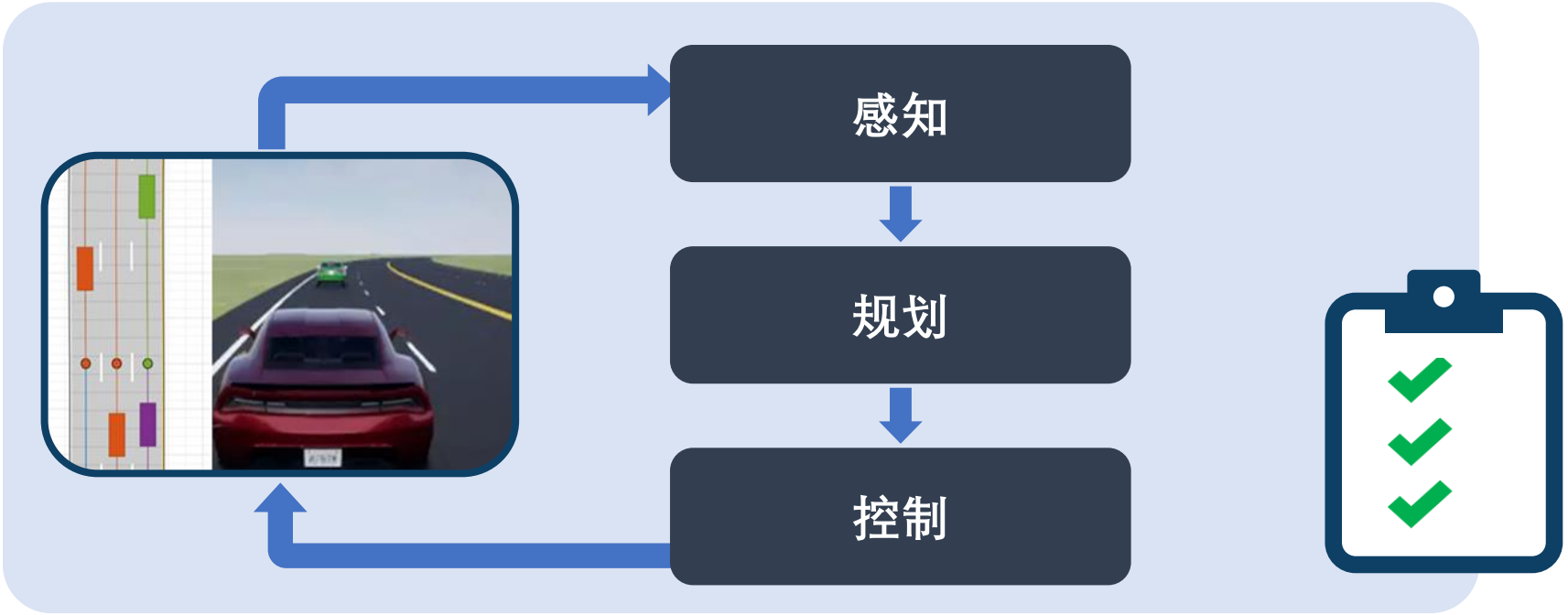
Simulink Test™

Simulink Requirements™

Simulink Coverage™

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我们可以帮助自动驾驶工程师...



如何
分析 & 合成
场景?

如何
设计 & 部署
算法?

如何
集成 & 测试
系统?